Android Application for Interior Designing of Smart Homes Using Augmented Reality

A Major Project Report Submitted
In partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology
in
Information Technology

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CERTIFICATE

This is to certify that this is the bonafide record of the project entitled "Android Application for Interior Designing of Smart Homes Using augmented Reality", submitted by S.Greeshma(18N31A12E5), U. Bhargavi(18N31A12G8), Saiyog Nagari(18N31A12E2), S. Jeevan Sai Chowdary(18N31A12E0) of B.Tech in the partial fulfillment of the requirements for the degree of Bachelor of Technology in Information Technology, Department of IT during the year 2021-2022. The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

I hereby declare that the project titled "Android Application for Interior Designing of Smart homes Using Augmented Reality" submitted to Malla Reddy College of Engineering and Technology (UGC Autonomous), affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH) for the award of the degree of Bachelor of Technology in Information Technology is a result of original research carried-out in this report. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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ABSTRACT

Decorating your home with the right items is not easy. Our project is to design an android application that will take a picture of user's room and allow visualizing it with different available interior designing options like carpets, curtains, wall paints etc.

AR technology is often used in applications that simulate an arrangement of furniture. We here will discuss system based on Augmented Reality for home exhibition, which supports real time tracking without identification markers; the system uses FAST corner detection. To solve this problem, this app we propose a method that can add virtual objects to the real time environment (AR) using camera. Augmented Reality furniture arrangement systems are useful for viewing room or building layouts without having to buy or move real furniture.

However, such systems often require users to physically and frequently change their viewpoint of the physical space, which requires manual manipulation of the same and is often limited to a 2D tablet or phone interface. To help address this problem, we have developed a system that automatically calculates the most suitable view point to improve understanding of the room layout as a whole, and allows the user easily transition to that viewpoint.

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1. INTRODUCTION

Augmented reality (AR) is a live direct or indirect view of a physical, real-world environment whose elements are augmented (or supplemented) by computer-generated sensory input such as sound, video, graphics, or GPS data. It is related to a more general concept called reality, in which a view of reality is modified (possibly even diminished rather than augmented) by a computer. As a result, technology functions by enhancing one's current perception of reality. Augmented reality brings out the components of the Digital world into our perceived Real-world. This project on Augmented Reality for home furniture has the potential to be a boon as an end-user application (a customer who wishes to buy furniture). It provides 3-D animated models (furniture models) and instructions integrated with the work domain in real-time. It enables the user to view and manipulate desired furniture and other objects usually used for interior home designing without visiting a furniture store.

Augmented Reality in simple terms can be stated as:

Computer/mobile generated data (Augment) + Physical/Real-world Environment (Reality)

Augmented Reality (AR) is a computer technique to harmonize the virtual world with the real world and is a hot topic of many researchers in recent years. Based on computer graphics and related image processing techniques, AR places virtual objects into the real scene and enables users a realistic immersion. Different from Virtual Reality (VR), AR emphasizes the application of the real scene to the virtual world. It picks a real-world scene and adds a virtual object to it thus the viewer sees the digital image as if it's real. There are two methods to match the virtual objects and real objects in AR technology: marker systems tack and markerless systems. The former takes LED, identification plate, etc. as the Camera tracking target; the latter takes the edge, plane, feature invariants, etc. complicated natural features as the camera tracking target. The greatest challenges in the implementation of MAR are tracking and registration. Because of the limited computing capability of mobile devices, the implementation of lightweight tracking and registration is especially important. To those markers less tracking system, applicability in the mobile devices should be noticed.

1.1 Purpose and Objectives:

Purpose:

Nowadays, people usually will go to the shop for samples to decorate house and room. But the majority of the people are busy with work. People are no time to go to various stores to buy furniture for their home, shop, or office. It is also difficult to let alone take it out and try it at home. If found out that the furniture is not suitable for the house, then the furniture needs to return to the furniture shop. Even if have time to go to the furniture shop, after a visit to the shop, people may find out that there is none of the furniture that fulfills the requirement. Hence there should have a way to let people use their mobile to view a 3D interior design in any way and anytime before it populates with the real items.

Next, that is difficult to fulfill the customer's contentment of decorating the room without an imaginary picture to refer will make them confused. And that is so hard to choose the sofa that is suitable for the house at the furniture shop. Normally, people will ask the opinion or suggestion from the staff of the furniture shop when can't decide which furniture suits the house. Hence, the staff of the furniture shop will recommend some of the furniture that may suitable. For the reason that there are limited spaces, so not all the furniture is displayed in the furniture shop. Thus, staff of the furniture shop may use words to express the design of the furniture that is not displayed in the shop. It is hard to explain in words, customers normally do not get a good picture of the design of the furniture because discuss without a visual design. It's wasting time and energy.

Other than that, there are a lot of people who throw off the furniture catalog. There are a few reasons that people act like that. Many catalogs that are obtained from the public are 3 not attractive enough. Although some catalogs have a nice and colorful design, the picture printed on the catalog is stark and lifeless, so it was difficult to attract the user's attention. Other than that, many catalogs contain a lot of descriptions, users are not willing to waste their time reading a catalog that contains a lot of unnecessary information. It is boring to look at this type of furniture catalog. On the other hand, in the industrial design field, some of the designers have used sketches with pens on paper to explore and evaluate early concepts. This is usually drawn in 2D or 3D hand sketches. In

In this case, if the interior design furniture is present in the hand sketch, a designer often encounters situations in which the client can't view better graphics of visualization. For example, the client can't view the design of the table, chair, cabinet, etc more effectively.

Hence, here is the purpose of building the project which is designing a mobile application that runs on Augmented Reality:

- Build a handy and easy-to-understand user interface to help the user analyze the furniture in the required space.
- Help them analyze costs from various stores and buy the best furniture within their budget.
- Navigate through various stores in the market out there so that he could compare different items of the same model and purchase the best one according to his requirements.
- Help the user to save unnecessary costs they spend on interior design professionals to make their home look smart.
- Help the user to save time navigating through a furniture store only to conclude the furniture he brought is not worth the money they spent.

Objective:

Currently, augmented reality technology is not well known in the market of India. But AR technologies would potentially benefit design modifications and refinements due to the support for viewing the design. Nowadays, AR technology has been brought into the field of education, entertainment, commerce, military, etc. If you have played PokemonGo, then you are already familiar with the concept, except the app deals with furniture instead of virtual Charizard. Therefore, the main objective of this project is to create an interior design application using augmented reality which to benefit people. It is to enhance the traditional catalog which is only printed with 2D images. A user just needs to download the application and install it on their device. Then play with the 3D virtual furniture in the house spaces. By having an interior design application, users can view the front view, side view, and back view of the virtual furniture and at the same time interact with the virtual model.

The next objective is to provide the user with a new interactive technology to be practiced in interior design. It is so interesting to enhance the perception of and interaction with the real world. Three types of gestures have been included in this application which are a drag, pinch, and rotation gesture. And the user is allowed to view the 3D virtual model of the furniture in real-time. Users just need to install the application on the device and start to design the ideal environment with several 3D virtual furniture models. Then choose furniture from the list and the model will appear on the screen of the device. Users can drag, rotate, enlarge and minimize the 3D furniture model to fit the model in the house. This technology has great potential to reduce costs, shorten design time for decorating the home space or room.

The other objective is to develop an augmented reality mobile application in interior design that is more convenient to the user. By having this augmented reality application, user can use their mobile to view a 3D interior design at anyway any anytime before it is populated with the real items. That is the point of interior design using augmented reality application in Android was recommended, which is providing a more precise, power-driven, and easier platform for users to choose the furniture that suits the room or house, without wasting their time and energy to go to the various store. User no needs to ask the advice from the expert on how to decorate the house because the user can decorate the room or house on their phones by using the application.

Last but not least, an augmented reality application for interior design is developed to produce a realistic virtual furniture model in interior design. It gives the feeling of realism. That is the 3D virtual furniture will look similar to the real furniture model. Therefore, by looking at the 3D virtual model that appears on the screen, the user will have a feeling of realism. And can decide which furniture is suited to their house.

Therefore, the objectives of the research are:

- To create an interior design application using augmented reality technology.
- To provide users with a new interactive technology to be practiced in interior design.
- To develop an augmented reality mobile application in 3D interior design that is more convenient to the user.
- To produce a realistic virtual furniture model in an interior design similar to real furniture.

1.2 Existing System and Proposed System

Existing system:

There is nothing worse than taking a day trip to a furniture store and assembling a piece of furniture only to find that it doesn't quite fit where you hoped it would go or that it doesn't match with the rest of the room. Or there could be another way which is quite expensive to hire a professional designer. Though many people opt for these two options, one is found to be quite expensive in the aspect of time and the other in the aspect of money. To overcome these flaws the Swedish furniture company, IKEA designed an app that runs on Augmented Reality. Though it is a good idea to be developed, it runs only on the latest version of iOS. One cannot download and use it on Android mobiles. Here are some disadvantages of using the existing system.

Disadvantages of manually designing the smart home:

- Purchasing furniture from a store has a bigger list of disadvantages. One can find only limited options as it is not practical for any store to store a wide range of products due to space constraints.
- It also consumes so much time to walk through the store and assemble the required products for the space. After doing so much if it doesn't suit the space, it is waste of time and effort.
- When designing the home manually it requires a professional to guide as it is not practical for an ordinary person to design it smartly. Hiring an interior design professional comes with a lot of disadvantages.
- Truth be told, hiring a professional designer is simply a luxury that not everyone can afford. Of course, many would argue that hiring a designer ends up paying for itself because it saves you from making costly design mistakes. On the other hand, if you enjoy designing a home and are capable of doing it yourself, then you will likely save plenty of money by not hiring a design professional.
- From paint colors and countertop materials to light fixtures and kitchen finishes, handing over all of the design decisions to an interior designer could make your home feel less personal. After all, if you desire for your home to be a place of self-expression – one that

reflects your personality and life experiences – then it might be easier for you to design it yourself.

- When hiring an interior designer, you must hire someone who shares your tastes and design aesthetics. Whatever your preference, make sure you share the same vision as your interior designer. Otherwise, you'll end up with a home you hate.

Disadvantages of existing app:

- Swedish giant in the furniture industry has put its technology forward to design an app that works using Augmented Reality. Though this app works wonders it has its disadvantages as follows.
- The app is designed for iOS devices and the latest versions only. It cannot be downloaded and used on Android devices since it is not available in the Play store. Since a lot of mobile users use Android, this app is not available to everyone.
- When this app is opened, it comes with a mandatory chatbot that has a predefined set of questions. Since the main objective of a user-friendly app is to make it accessible to every user out there, not all people can answer the chatbot.
- It also has a pre-defined set of items from the IKEA store. Users cannot use any other product to review it which is the main objective of this app. It doesn't have the facility to sign up and save the content for later use. So, it is not customizable from person to person.
- It cannot tell the quality of the furniture. It doesn't have a clear user interface. It has few icons and animations that confuse the user. Hence, it is difficult to use. Also, objects are too hard to control.

Proposed System:

This application augments a 3D model of the desired furniture on the current live camera stream in real-time. Thus, it helps the customer who wishes to buy that particular piece of furniture can virtually see that furniture model in real-time and can interact with the system to fit perfectly. The user gets to decide whether he/she wishes to buy that particular piece or to choose a piece that is more suitable to his/her requirements. The proposed solution helps the user to get an approximate view of the furniture in the desired position.

Advantages of the proposed system:

- This is an application that is automated and portable which can be easily handled by almost every person out there. The main objective of this application is to make it handy to solve complex interior design tasks.
- This application can be downloaded and used on Android devices. Android is the most commonly used operating system in mobile devices these days. Hence, this application can reach every small individual trying his best to save the best they spend on smart home interiors.
- This application doesn't have a mandatory chatbot which seems to be a lot of work when people with low technical knowledge use it to search for their requirements. Instead, it has an options menu which is a lot easier to handle than a chatbot.
- Instead of having items from a single store, it has items from different stores and of different varieties, so that the individual can browse on his approximate choice and see if it fits in his space which can be quite affordable sometimes.
- Not all can afford the furniture from costly stores. The main objective is to reduce the indispensable costs people put in interiors. Hence, the user should have the facility to navigate through different stores and compare prices.
- It has a signup page too which is not mandatory for the user to login. But when signed up, the user can create his customized environment to save his goods or wish to list them and visit the link provided to purchase the item within the budget.
- Today's trend is cheap and better aesthetics furniture. The key factors in this decision-making are price, quality, durability, and functionality. There is also a rising demand for online purchased goods. Manufacturers and interior designers become more creditable. As well as the impact on personal information sources is greater on furniture purchase decisions. So, considering all this our application integrating Augmented Reality can match the current trend of users

1.3 Scope of the project

- **User:** Everybody that has a mobile gadget to use the augmented reality application to design their room or house with virtual 3D objects.
- **Furnishing companies:** Implement augmented reality applications with virtual 3D representation furniture and furnishings for their customer.
- **Content application:** The application is an interactive AR application. The model focus is the furniture of the living room. So, the 3D models that develop will be the furniture of the living room. To use this application, the user needs to have the camera on their device. After that, the user just needs to install the application, turn on the application and point the camera to the living room. Users can take a picture of the design space and set the image as a tracking pattern. Then, the user can choose the furniture from the list and the sofa's image will be overlaid on the room. The whole process can then be reset by reactivating the camera.
- **Interaction technique**: Three types of gestures have been provided in this application such as drag, pinch, and rotation gestures. Users can drag, rotate, enlarge and minimize the furniture model to fit in their house. At the same time, the user can view different dimensions of the 3D virtual object such as the front view, side view, and back view of the model by rotating the model with fingers.

2. LITERATURE SURVEY

Phan & Choo(2010) states that few previous authors had proposed interior design applications by using Augmented Reality (AR). They are authors Koller, C. Woodward, A. Petrovski; K. Hirokazu, et al. The related equipment such as data glasses connected Head-mounted display (HMD). Additionally, various lightweight solutions using a PDA device have been recommended by the AR Team in Findland (S. Sitanen and C. Woodward, 2003). Nevertheless, this equipment is uncommonly used by users.

1. Interior Design on Augmented Reality Environment:

Abstract:

This article presents an application of Augmented Reality technology for interior design. Plus, an Educational Interior Design Project is reviewed. Along with the dramatic progress of digital technology, virtual information techniques are also required for architectural projects. Thus, the new technology of Augmented Reality offers many advantages for digital architectural design and construction fields. AR is also being considered as a new design approach for interior design. In an AR environment, the virtual furniture can be displayed and modified in real-time on the screen, allowing the user to have an interactive experience with the virtual furniture in a real-world environment. Here, the AR environment is exploited as the new working environment for architects in architectural design works, and then they can do their work conveniently as such collaborative discussion through AR environment. Finally, this study proposes a new method for applying AR technology to interior design work, where a user can view virtual furniture and communicate with 3D virtual furniture data using a dynamic and flexible user interface. Plus, all the properties of the virtual furniture can be adjusted using occlusion- a based interaction method for a Tangible Augmented Reality.

2. Usage of the powerful technology stack for interior design:

Abstract:

This project provides the public with an attractive and interactive interior design application using Augmented Reality. Nowadays, there are a lot of people familiar with computer system and communication technologies. Hence this exploring the prospect of a new form of Virtual Reality (VR) application called Augmented Reality (AR) technology develops more rapidly. Augmented reality is a field of research which deals with the combination of real-world and virtual environment. It is an environment that applies between virtual reality and real environment. There are several problems induce me to create an interior design application in AR. Nowadays, people are busy with their work. People are no time to go to various stores to buy furniture for their home, shop or office. There is difficult to fulfil the customer's contentment of decorate their room without imaginary picture to refer.

Printed furniture catalogue is basically a paper-based information with lots of texts and images which does not provide any interaction for the user. And people can't view a better graphics of visualization because the design layout is statically presented on papers. Due to that, AR and 3D technology must be applied into interior design area. It is an application which includes furniture and furnishings will develop in three-dimensional surface of object and present by using augmented reality. It will perform the interactive and attract the attention from the public. In this project, I choose Waterfall methodology as a guideline to develop my product throughout this project. By having a methodology, I can carry out my project successfully. As we know, augmented reality technology is still new in our country, but it is a fact that AR technologies would potentially to be the future technology trend to the public.

3. Augmented Reality uses in Interior Design:

Abstract:

Increasingly, interior design retailers rely on online stores to reach their customers as more and more people are connected to the internet. One of the issue in online shopping is that customers cannot visualize how new furniture pieces would blend in their home, but that is not the case anymore as Augmented Reality (AR) becomes more mainstream. This paper analyses several AR applications used in the interior design and highlights their pros and cons. Based on this analysis, we propose an AR application that solves most of the current issues of the interior spaces design.

4. Augmented Reality in Home design:

Abstract:

This research is about the application of Augmented Reality (AR) in interior design education. AR has been a widely researched technology in many fields, and education is one of them, it is also regularly used in civil engineering, architecture, and interior design research. For interior designers, being able to think in three dimensions and visualizing projects is of great importance. The purpose of this study is to analyse the effect of using AR in design process to find the optimum space arrangement under the following criteria: the perception of 3D and 2D space, the visualization and analysis of the given space, and the usability of AR on mobile devices. For this research, a group of third grade interior design students in Arel University, who had already attended a project lecture, were selected. These 14 students, who were familiar with our traditional education process, were asked to use AR in their next project to design a specific area that they were not able to physically visit. These students were assigned with creating this space from instructions and measurements by computer-aided modelling (CAD) and start the design by using augmented reality with their mobile phones when they needed to. After their work finished, data was obtained via a questionnaire survey. The survey included the students' feedback on their learning experience, whether AR had been advantageous during the design process, whether the

students had experienced problems with the interface, identification of the process where they preferred to use AR and if the learning method was motivating. This study is expected to contribute to knowledge on the advancement of Augmented Reality (AR) applicable to design education.

IKEA, a worldwide known “Assemble & Install-It-Yourself” furniture company with Swedish origin, launched an augmented reality app, namely, IKEA Place, that aimed to solve practical problems surrounding furniture shopping in 2017. The IKEA Place, which used augmented reality to allow its users to visualize how furniture will look in their own home, is examined in this article.

5. Furniture Layout Application using Augmented Reality:

Abstract:

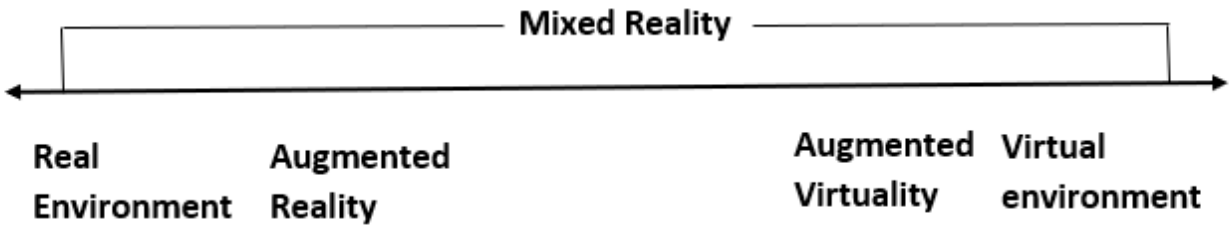
Information technology currently supports the development of human interaction with virtual environment. This development will continue in developing in the form of Human Computer Interaction (HCI). Augmented Reality is a field of computer research which deals with the combination of reality with computer generated data. The main purpose of this project is to develop a windows application for trying different furniture items in a virtual way. This application will eliminate the need of physically visiting the furniture store which is very time consuming activity. This research is using photo-realistic 3D models to rendering used in computer graphics. Rendering of the basis for colour and shading in order to make it appear solid and 3D. These 3D furniture models are superimposed on to the live feed of real space taken from the camera. After imposing the model it will appear as, it is actually placed into the real world. Using this application one can choose from the different set of furniture and try it on their space.

Inputs Taken:

In an AR environment, User can play with the virtual furniture that displayed in real-time on the screen. It provides a platform that users can interact with the virtual object in a real space. Hence, this research suggests a new way to apply AR technology in the interior design work. With

dynamic and flexible user interface, user can look at the virtual object and correspond with the 3D virtual object.

For the reason of Augmented Reality (AR) covering of computer graphics on the real world, this make it became a novel technology. AR is also termed Mixed Reality (MR), mixed reality (MR) which encompassing both augmented reality and augmented virtuality.



From other research paper, Charles and Sanni have experimented Augmented Interiors with Digital Camera Images. Based on the paper, they develop a system for Augmented Reality interior design based on digital images. With just PC, digital camera and printer, they successfully introduce a solution for augmented interior design. Those are the things that can simply get. The marker can place on anyway of the floor of the space of the house. It will determine the scale and coordinate of the space of the house after the marker has been place. This provides the opportunity to let user to choose the virtual furniture that they want to be placed into the spaces. Actually, similar system has earlier been presented, but the specialities of this system are added more functionality to the user interface and improved implementation issues. For example, there is ability to manage objects and adding lighting and shadows to increase the level of realities. They also point out the important of removing the marker from the final images.

The other research paper, Tiina and Sanni (2012) describe a process of the collaborative design and fulfilment request of an interactive interior design service system. The technological purpose was to create interior design concepts that utilize augmented reality (AR), 3D models and user-generated content within the system framework. They come out with the important AR features of the interior design service such as realistic lighting, and number of variety furniture models available. By using the AR technology, it is important to let user visualise a lighting that are

more real for the whole day weather based on the location of the windows. This can provide user more realistic light when use the system. With the 3D representation of interior design, user also can't view a better graphics of visualization. For example, User can view the design of the table, chair, cabinet and etc more effectively. Finally, it was important to let user or designer to sense that decorate the rooms and establish a plan was interesting and pleasure when using the interactive interior design service system.

In the future, it believed that many developers will seek augmented reality interaction more deeply, providing interior designer, Furniture Company and etc with more function augmented reality design tools. It also let user interact with a virtual object with simple and easy to use tracked physical tools.

Markers in Augmented Reality:

By definition, augmented reality provides users with an opportunity to integrate virtual 3D objects into their real-life environment in real-time. But, there are different ways to do so, including marker-based, marker-less and GPS-based solutions. Each has its benefits and drawbacks.

Augmented reality presents a growing market and it is expected that by 2025 it will grow to more than 198 billion U.S. dollars (From 3.5 in 2017). Although the sector has huge potential, it is currently at a stage where the technology both in terms of hardware and software is undergoing a massive revolution. Currently, augmented reality is accessible via mobile devices and three distinct experiences are offered: marker-based, marker-less and location-based. All of those have certain limitations and developers are working on eliminating years in the coming years. Here I look at the present state, bet there are improvements to all challenges coming up.

- **Marker-based augmented reality:**

Marker-based augmented reality experiences require a static image also referred to as a trigger photo that a person can scan using their mobile device via an augmented reality app. The mobile scan will trigger the additional content (video, animation, 3D or other) prepared in advance to appear on top of the marker. Marker recognition can be local or cloud-based, it means that

marker databases can be stored on device and recognition also happens on device. The databases can also be stored on a cloud and recognition happens on a server; phone is only sending point clouds to server. Device-based recognition can happen immediately, but if cloud recognition is used, then it will take a while longer for the content to be downloaded from the server. Usually, it takes a couple of seconds before the user can see any augmented reality experience.

The trigger image (marker) has to be unique, avoid using stock photos by all means (if you are using apps which are cloud record based) as these may already be utilized by other projects or apps. It's better to capture your unique content or create a custom-made design to avoid such issues. To bring this unique photo to life and show your augmented reality content, the mobile device regardless of the app you're using, will use computer vision.

Advantages:

- If the marker image is prepared correctly, marker-based AR content provides quality experiences and tracking is very stable, the AR content doesn't shake
- Easy to use, detailed instructions are not required for people who use it for the first time

Disadvantages:

- When the mobile camera is moved away from the marker, AR experience disappears and the trigger photo has to be scanned again. It is possible to use extended tracking, but in most cases, extended tracking makes things worse.
- Scanning will not work if markers reflect light in certain situations (can be challenging with large format OD banners in ever-changing weather conditions)
- Marker has to have strong borders/contrast between black and white colours to make tracking more stable. Smooth colour transition will make recognition impossible.

❖ **Marker-less augmented reality:**

Marker-less augmented reality works by scanning the surrounding environment and there is no trigger photo necessary to retrieve the augmented reality content. Apps that offer such features usually will ask the user to find a flat surface such as a table or floor for placing the AR elements as the objects will not always make sense floating in the air. For computer vision to

detect the flat surface, it has to be textured. You will find it challenging or even impossible to use on a white background or on other single colour surfaces.

Advantages:

- Once the content is placed in a room, it is more flexible than marker-based alternatives.

Disadvantages:

- The augmented reality content may not make sense in a certain context.
- For better experience, it is required that the surface has a texture for computer vision to recognize it.

A cool example of marker-less content was designed by the TIME publication to create the Apollo 11 'Landing on the Moon' experience.

❖ **GPS-based augmented reality:**

GPS or location-based AR solutions respond to the sensors of your mobile device. It allows for objects to be placed in a certain location and as long as the public has the app, they can retrieve it regardless of the time of day or weather conditions.

An example of this is SAN.app, where its author, Gints Gabrāns, uses solely GPS to place his augmented reality artworks around the globe. Another example, which addresses the challenges of GPS sensors whilst still offering location-based AR experience is the brand-new Google solution for walking. One thing to note in the latter example is that Google asks the user to scan their surroundings and finds itself based on urban static objects aka buildings not solely on GPS.

Advantages:

- Allows for geographical targeting in tourist hotspots without requiring expensive outdoor banners.
- Allows for practical applications in terms of directions.

Disadvantages:

- It is challenging to get precise GEO-located experiences on mobile devices because of lack of sensors accuracy in phones.

3. SYSTEM ANALYSIS

2.1 Hardware and Software Requirements

Hardware Requirements:

- **Processor and Speed:** 64-bit, four-core, 2.5 GHz minimum per core.
- **RAM capacity**: 4 GB for development and evaluation use.
- **Hard Disk**: 10GB for development and evaluation use in a total capacity of 80GB.

Software Requirements:

- **Operating System:** Windows and Android.
- **SDK IDE:** Android Studio.
- **Virtual Emulator:** Nexus Pixel 2
- **Programming Language:** Java, XML
- **Tool:** ARCore with Sceneform

2.2 Software Requirements Specification:

Input Design:

In making of this application, it requires two types of inputs. One is to upload the predefined images and the other is to ask the user to input the environment to fit the required furniture.

❖ **Input-1**

First kind of input is to store the images or 3D models into the storage source so that the user can access them. We have to input the 3D models to storage in the form of .glb files. GLB is a 3D file format that's used in Virtual Reality (VR), Augmented Reality (AR), games, and web applications because it supports motion and animation. Another advantage of the format is its small size and fast load times. GLB files are a binary version of the GL Transmission Format (glTF) file, which uses JSON (JavaScript Object Notation) encoding. So, supporting data (like textures, shaders, and

geometry/animation) is contained in a single file. There are 2 ways to grab a 3d model; download from web or make one yourself. If you want to download it you can go to an awesome 3D model repository by Google called as Poly and search for any .glb file and download it for free. You can also use a 3D computer graphics software to make one yourself. One of such software is Blender.

Blender Software:

Blender is the free and open-source 3D creation suite. It supports entirety of the 3D pipeline- modelling, rigging, animation, simulation, rendering, compositing and motion tracking, even video editing and game creation. Advanced users employ Blender's API for Python scripting to customize the application and write specialised tools; often these are included in Blender's future releases. Blender is well suited to individuals and small studios who benefit from its unified pipeline and responsive development process. Examples from many Blender-based projects are available in the showcase.

Blender is cross-platform and runs equally well on Linux, Windows and Macintosh computers. Its interfaces use OpenGL to provide a consistent experience. To confirm specific compatibility, the list of supported platforms indicates those regularly tested by the development team.

As a community-driven project under the GNU General Public License (GPL), the public is empowered to make small and large chances to the code base, which leads to new features, responsive bug fixes, and better usability. Blender has no price tag, but you can invest, participate and help to advance a powerful collaborative tool: Blender is your own 3D software.

We can store these files in the databases or in the cloud based on the requirements of the user. Here the primary actor is the developer. When the user requires more inputs to be added then the developer has to fulfil the requirement.

.glb Files:

GLB is a 3D file format that's used in Virtual Reality (VR), Augmented Reality (AR), games, and web applications because it supports motion and animation. Another advantage of the format is its small size and fast load times. GLB files are a binary version of the GL Transmission Format (glTF) file, which uses JSON (JavaScript Object Notation) encoding. So, supporting data (like textures, shaders, and geometry/animation) is contained in a single file.

The Khronos Group developed the GLB and glTF formats in 2015. They saw a need for formats that developers could open and edit in many graphics and 3D apps. Version 2.0 of the specification (released in 2017), added Physically Based Rendering (PBR), which allows shadows and light to appear more realistic and coding updates for speed and improvements in animation.

GLB, like glTF, is a royalty-free way to encode 3D data. A GLB file will generally be smaller (approximately 33 percent) than a glTF file and its supporting files. It's also easier to upload a single file to a server rather than two or three, so GLB has an advantage there too. Many developers find it easier to edit the objects and senses in the glTF format and then convert it to GLB to deliver the file.

Khronos posted the specifications for glTF versions 1.0 and 2.0 on GitHub. The GLB specs are in the glTF documents. GLB files contain a header that identifies the file as GLB, the specification version, and the file length; a JSON chunk with the data included in a glTF file; and binary buffer, which contains the data included in the supporting files.

Derived from JavaScript, JSON is an open-source format used for data interchange and is usable by many programming languages. JSON's data types include arrays (an ordered list of values of any other type), objects (non-ordered collection of name and value pairs, the names are strings), strings (any set of Unicode characters, which can also be empty), numbers, Boolean values (true or false), and null (any empty value). Using these, a developer or an application can represent many items from contact information for a person to a 3D object in a GLB file.

glTF uses URIs (Universal Resource Identifier), which may also be called data URLs. This bit of code allows a developer to encode objects (from text to an embedded 3D object) in a web page. In glTF files, URIs pull data from the supporting files. Since the supporting file data used by glTF files are stored in the GLB files, URIs are not required. Therefore, GLB files are faster to load and process than glTF, making them useful for mobile- and web-based 3D modelling applications.

GLB files are useful in scenarios that require animation and motion, or when there is a desire for interactivity. Those situations include:

- Video game developers can save their work as GLB files.
- Interactive websites and ads use the GLB format to display objects that users can interact with (like a couch that they can rotate and view from any angle).
- AR and VR developers can also save their work in the format.

❖ **Input-2**

Next type of input is capturing the environment in order to fit the required furniture at required location. This can be done in different stages. At first the app is opened and the user has to login or signup with the information. User can also skip this step. This information is stored in local DB or in the cloud platform. After login process is done one can select the required categories and can go to the camera option in order to scan the environment. Preferably Capture the image in with a plain background like wall, which does not have many colours or texture. This will reduce the time required for marker detection. Next, we have to add the markers for detection. An augmented reality marker is an image or an object that can be recognized by an AR-enabled mobile app and is used to trigger augmented reality features. When designing your marker image, here are other things to take into account:

- Using graphic elements repeatedly will make the marker look the same from all angles. If it is confusing to the human eye, it will be even more consuming for computer vision to detect which direction the marker is placed and retrieve it. Choose irregular shapes and photos that look different from all angles, so the computer vision knows if it is upside down per se.
- Do not use logos as marker images. As tempting as this may seem, a logo may appear in multiple materials in different contexts. If the logo is a marker, it will confuse computer vision. The same goes for the Overly logo. It may be present in other designs, so do not include it in the marker. This will ensure that the right content comes to life when your target is scanned. However, it is good to place your AR provider's logo within the instructions so people can quickly see what app has to be downloaded.
- Similarly, to logos, avoid using the same images or symbols within your design. You can keep your design style aligned while ensuring all elements are unique. For the computer vision algorithm to recognize a marker image, it needs to recognize just around 70 per cent of the marker's feature points. If key symbols are replicated, and only some elements differ, it will be impossible for the algorithm to differentiate your designs and retrieve the correct content.
- The excellent aspect of this feature is that if an image gets distorted by reflections or light once printed and placed in the desired location, as long as the computer vision can detect most of the feature points, your AR experience will still get retrieved.

- Computer vision doesn't understand blurred images, so skip on blurry backgrounds whenever possible. Aim for high-resolution images with distinct textures. Remember—blur has no feature points.
- When people scan your AR marker with their mobile device, it must take up at least 50% of their camera screen—no less. So, your AR marker could not be a slim line or a tiny dot. Stick to minimal whitespace. You will once more have to think about the marker image and its size once your project goes to print.

The user has to wait until the camera detects the environment and creates a marker or some dimensions to fit into it so that the user can adjust the furniture. The primary actor here is the developer and camera where the 2D black and white markers are added into the database.

Output Design:

According to two types of inputs given, there are types of outputs. When the user logs in to the app, the system has to output the successful and unsuccessful login with the user's customized environment. Another kind of output is the display of input space with the best fit furniture from the selected category. It displays the 3D view of the furniture with which the user can interact with. The app provides the user with different modes to operate with the output. The user can be able to pinch in and out, rotate, drag using fingers. It makes the visual more convenient for the user to analyse according to the requirements. He will also be able to wish list it in order to purchase it from the shopping site. This makes it more convenient for the use to interact with the user interface. If the user doesn't get satisfied with the output shown, he can again select the category from the menu provided and get the furniture visuals to analyse. After the process is done, he can return to the home screen by clicking the button in the UI. Here the primary actor is augmented reality manager.

Objective:

Building a home is a dream for every individual. A beautiful home comes with beautiful interiors which leaves some costly bills behind. It requires a lot of effort, money, time, and manpower. The cost of building homes cannot be avoided as it requires time and effort. But when given a thought one can reduce the cost of interiors. Growing technology brings Augmented Reality close to ordinary people and helps them save unnecessary bills. With this, an automatic android application can be designed which helps people work on interiors without any kind of help from interior designer professionals. The main objective of this application is to use a powerful technology stack to reduce efforts, time, money for users so that they can use it on something really useful.

- The objective of building this project is to provide a free and open-source android application capable of delivering accurate visuals of the best fit furniture to the space or location provided.
- It helps the user to navigate through different furniture at the ease of their home instead of spending a whole day running throughout the store just to realize the selected furniture doesn't fit the space.
- It also helps in reducing the cost to be paid to the interior design professional and helps in taking out the creative skills out from an ordinary person and help them make their home more personal and beautiful.
- The app works on the Android operating system which can be downloaded from the play store and installed it on the system. After installing it, it can be used as any other apps out there.
- The user when opens the app, it asks the user to sign-up or login if he has the account already. After signed up, the user information is stored in the local DB or in the cloud platform. This signing up stage can be skipped, but the user can get the best customized experience if signed in.
- The user is then asked to selected the type of room for the furniture to fit in and the type of furniture and later the model of the selected furniture.

- After selecting the category, the user is allowed to open the camera and is asked to show the location where the furniture has to fit in. Move the phone to different directions until the markers and dimensions appear.
- Later, the system displays the selected furniture within the input environment and helps the user to interact with it using different gestures like pinch in and out, rotate, drag and wish list it if satisfied.
- **Non-Functional Requirements:** The non-functional requirements define how a system is supposed to be. In this project, they are:
    - Run on newer version of OS.
    - Make use of best practises to avoid data leak over Wi-Fi.
    - Do not reveal information to anyone not an authenticated user.

4. SYSTEM DESIGN

4.1 Description:

There are various modules in the design of this application. They are represented diagrammatically as follows:

Module-1 Capture Image Module

Primary Actor: Camera

In this module the image of the environment is captured using camera that is available in the mobile device. This module has the primary actor as camera. The goal is to capture live images which will be used for detection. The main step is to capture the image with a plain background like wall, which does not have many colours or textures. This will reduce the time required for marker detection.

Module-2 Add a Marker Module

Primary Actor: Developer

This module helps the user to augment new modules into the database. When user request for new models, the developer creates it hence making the developer as the primary actor. In case, if the end user wants new objects to be augmented, then the developer has to create a new marker. A new 2-D, black and white marker is created and added into the database of existing markets.

Module-3 Add a 3D Model Module

Primary Actor: Developer

In this module if the user wants new models to be augmented, the developer adds a new model into the database. Hence, developer is the primary actor. In case, if the end user wants new objects to be augmented, then the developer has to create a new model.

1) A new 3-D model, similar to the objects to be augmented, then the developer has to create a new model.

2) Add this new model in the database of existing model.

Module-4 Detect Marker in Environment Module

Primary Actor: Augmented Reality Manager

In this module, the system captures a live stream of images in order to compare them with pre-defined markers by separating the foreground and background. The primary actor here is augmented reality manager. The system captures a live stream of images in order to compare them with predefined markers.

1) The first step to detection process is to capture live image using the camera.

2) During this phase, we separate foreground from the background as we know that our marker will be present in the foreground.

3) In order to simplify the above procedure, we make use of black and white markers and thereby first perform image thresholding onto the captured image, to obtain a binary image.

Module-5 Capture marker image with database Module

Primary Actor: Augmented Reality Manager

In this module, the manager provides a Boolean result by checking if the item is present in the database or not. Here the primary actor is the Augmented Reality Manager. Manager must perform pattern matching to provide a Boolean result i.e. present or not.

1) The system first extracts the salient features of the image like corners, edges, curves etc. which will help in comparison with existing markers.

2) Based on comparison, if the match is above a particular threshold level then true is given as the output else false.

Module-6 Augment 3D model on the marker Module

Primary Actor: Augmented Reality Manager

In this module, the model identified with the associated marker is placed on that marker. Here the primary actor is Augmented Reality Manager. The model associated with the identified marker is placed on the marker.

1) Once the marker has been identified, system needs to calculate the location and the orientation of the identified marker.

2) As per the orientation computed, manipulate the models scale and rotation and place the model on computed location.

The figure below illustrates the working of the system clearly as follows:

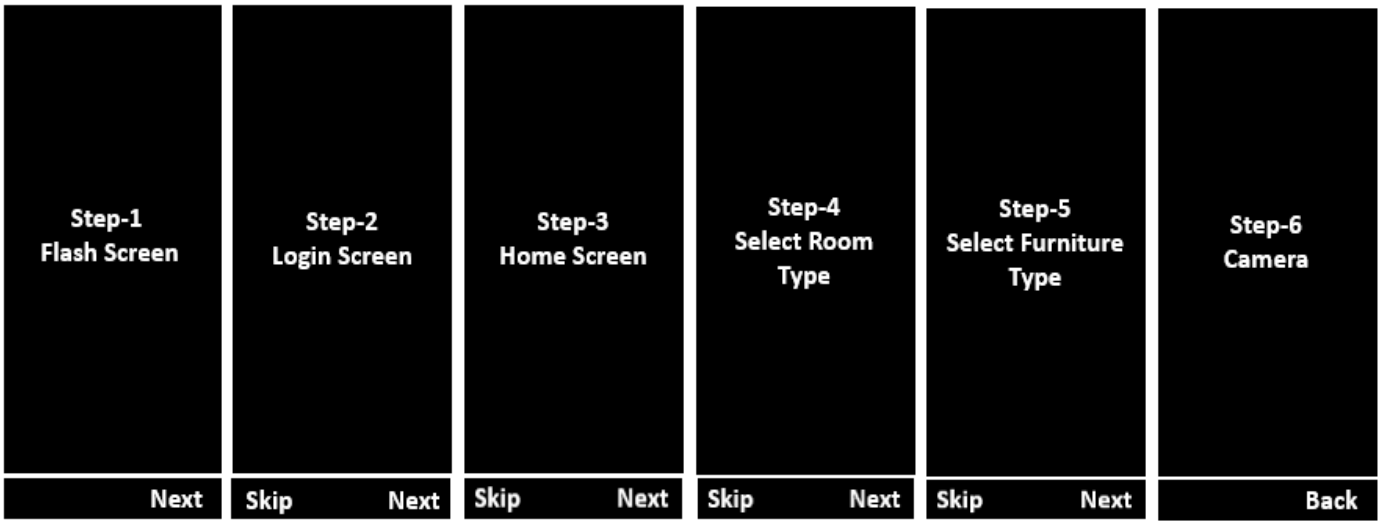


Fig 4.1.1 Working

Step-1: Click on the application which opens the flash screen showing the name of the app for a few fractions of seconds.

Step-2: A login screen appears which shows two options; skip and next. A user can skip and use the app without logging in. But he loses all his data after ending the app.

Step-3: After login, a home screen appears which has a selection menu to select the type of room in the house.

Step-4: After selecting the type of room, there is another selection menu to select the type of furniture to be checked in the selected room.

Step-5: After selecting the furniture type, camera opens. User has to show the environment to the app which marks the points and displays the selected furniture in the space captured.

4.2 Architecture:

- **Vuforia Architecture:**

Vuforia is a free software development kit for implementation of Mobile Augmented Reality. It is launched by Qualcomm in 2010, and the latest version is 2.8. With support for iOS, Android, and Unity 3D, the Vuforia platform allowed us to write a single native app that can reach the most users across the widest range of smartphones and tablets. Besides the basic function of Augmented Reality, Vuforia provided us various functions such as Text Recognition, Cloud Recognition, Multi-Targets, Frame Markers, Video Playback, Cylinder Recognition and Virtual Button Interactive etc.

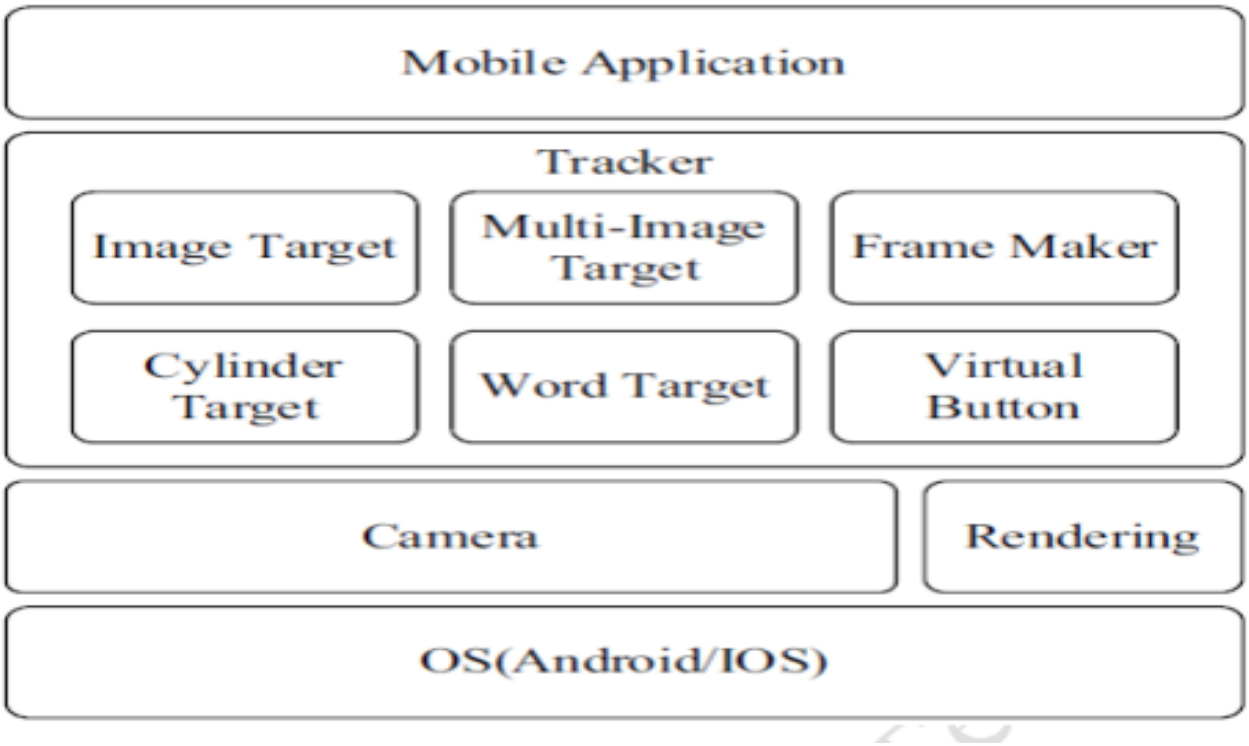


Fig 4.2.1: Vuforia Architecture

Two methods of storing image database are provided: device database and cloud database. The main difference between them is that the former does not require network connection, time

response is faster, is limited to 100 targets per downloaded device target database and has no metadata support whereas the later requires network connection and cloud recognition, time response is depends on network conditions, supports more target database (>1 million targets) and metadata. In this paper, device database is used.

The process of development with Vuforia can be divided into four stages:

- Create local image target database used for Augmented Reality.
- Define the configuration of image target in XML files.
- Load XML files and start image recognition in program.
- Load 3D models and display it.

On Android platform Augmented Reality mainly works on:

- Make static 3D models by 3D software.
- Choose suitable area in which collect images in target environment, generate target image on Vuforia platform, and then obtain target database files on Android platform.
- Recognize the target in image database by Vuforia, realize the control of 3D model with the help of Rajawali 3D framework.

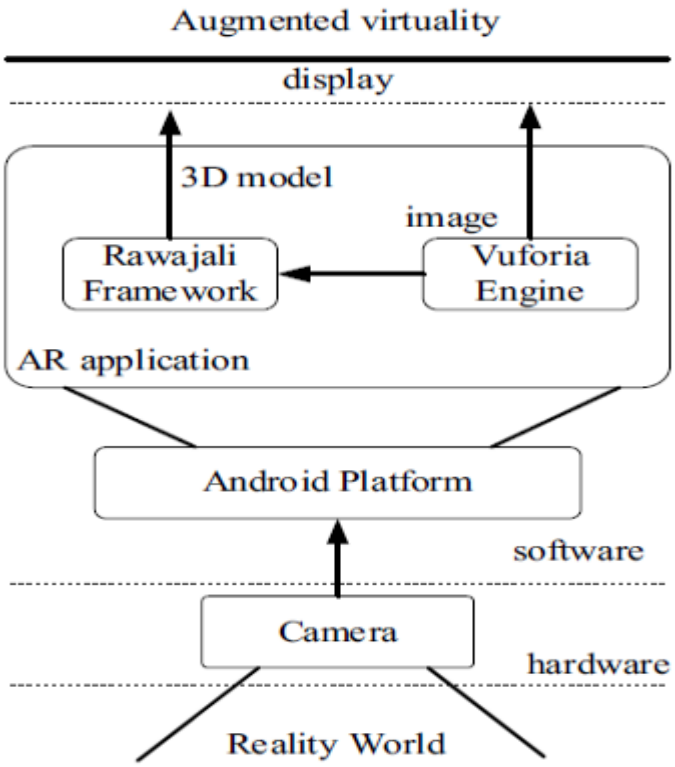


Fig 4.2.2: Application architecture on Android platform

- **System Architecture:**

The system architecture has different phases from giving and input to receiving an output. Input stage has two categories, receiving the user information through sign in page and inputting the 3D models into the local DB. It creates the markers and then stores in the storage environment. So, it works in two ways. After taking the input, it helps the user to select the category and helps in analysing the furniture at the specified location. The user is provided with the camera option which helps the user to scan or show the environment to the system. It then captures the environment provided by the user. Later the system checks for the markers that are suitable for the furniture and environment and separates the foreground to background. If match is found, the system stacks the selected furniture model on the marker. If match isn't found, the process starts again asking the user to select another type of furniture model. The diagram below explains the architecture of two different types of input.

Explanation:

Step-1: There are two types of inputs, the user data and 3D models or 2D markers. These data are recognised and stored in the data base or in the cloud platform.

Step-2: After the input data is given, the user is given with choices to select the category from the room, furniture category, furniture model.

Step-3: After the selection of the categories, the camera option opens, and the user needs to scan the environment, in where the Augmented Reality Manager recognises it.

Step-4: After this process, the foreground is separated from the background and the augmented reality manager searches for the marker that suits the selected model category.

Step-5: If the marker is found, it shows the dimensions on the screen and the 3D model is stacked upon the marker. If the marker is not found, it has different process gain.

Step-6: If the marker is not found, that means the selected category of 3D model is not present in the database. So, it is the duty of the developer to insert the 2D marker and 3D model into the database.

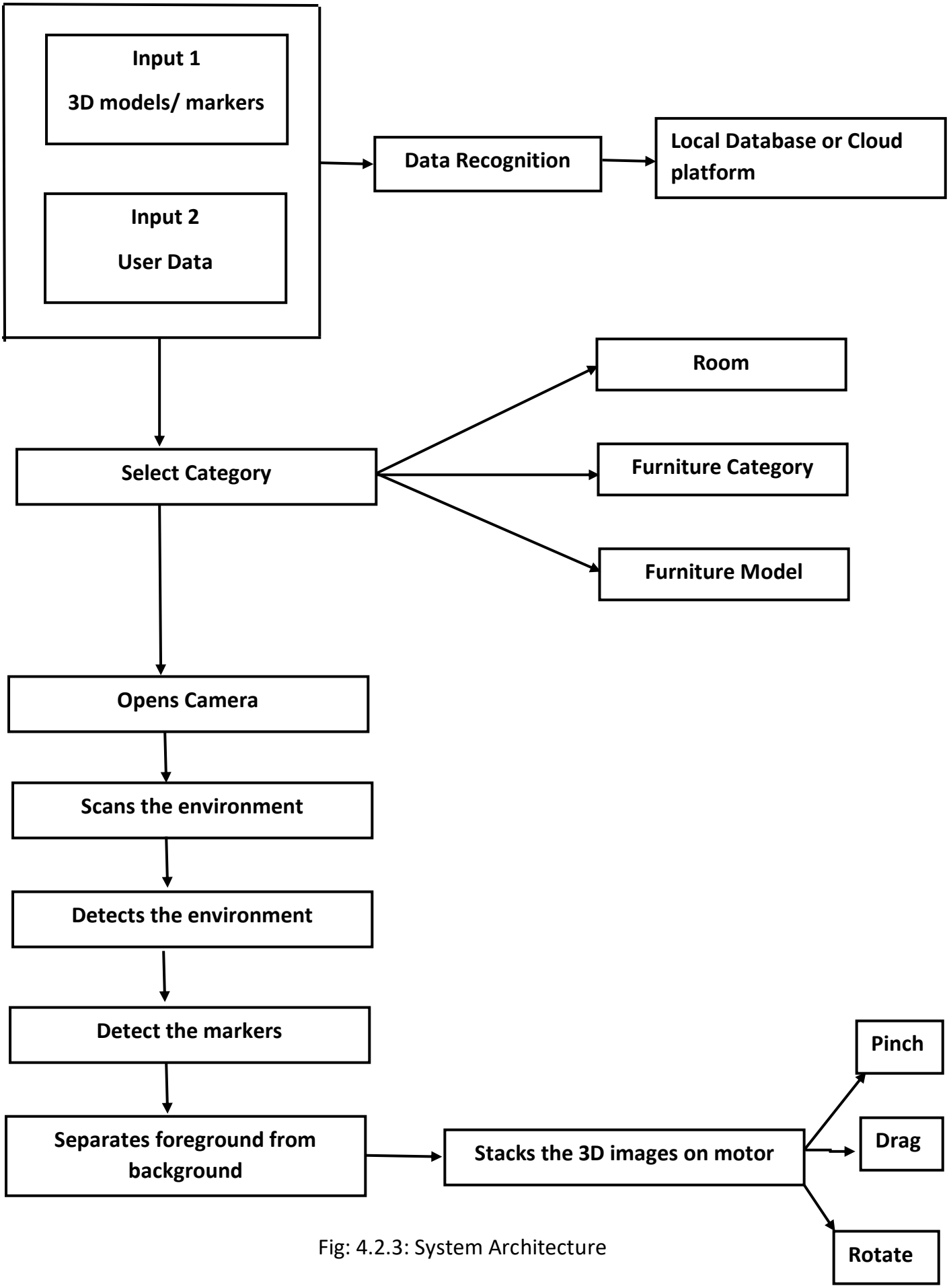


Fig: 4.2.3: System Architecture

4.3 UML Diagrams

- **Use Case Diagrams:**

Use case diagram is the primary form of system/software requirements for a new software program underdeveloped. Use cases specify the expected behaviour (what), and not the exact method of making it happen (how). Use cases once specified can be denoted both textual and visual representation (i.e. use case diagram). A key concept of use case modelling is that it helps us design a system from the end user's perspective. It is an effective technique for communicating system behaviour in the user's terms by specifying all externally visible system behaviour.

Actor	Tasks
Augmented Reality Manager	Detect the marker in the environment, Compare marker image with database, Register the background image, Augment 3D model on marker, Capture image, Display specification
Camera	Capture image
Developer	Add marker, Update database, Add 3D model

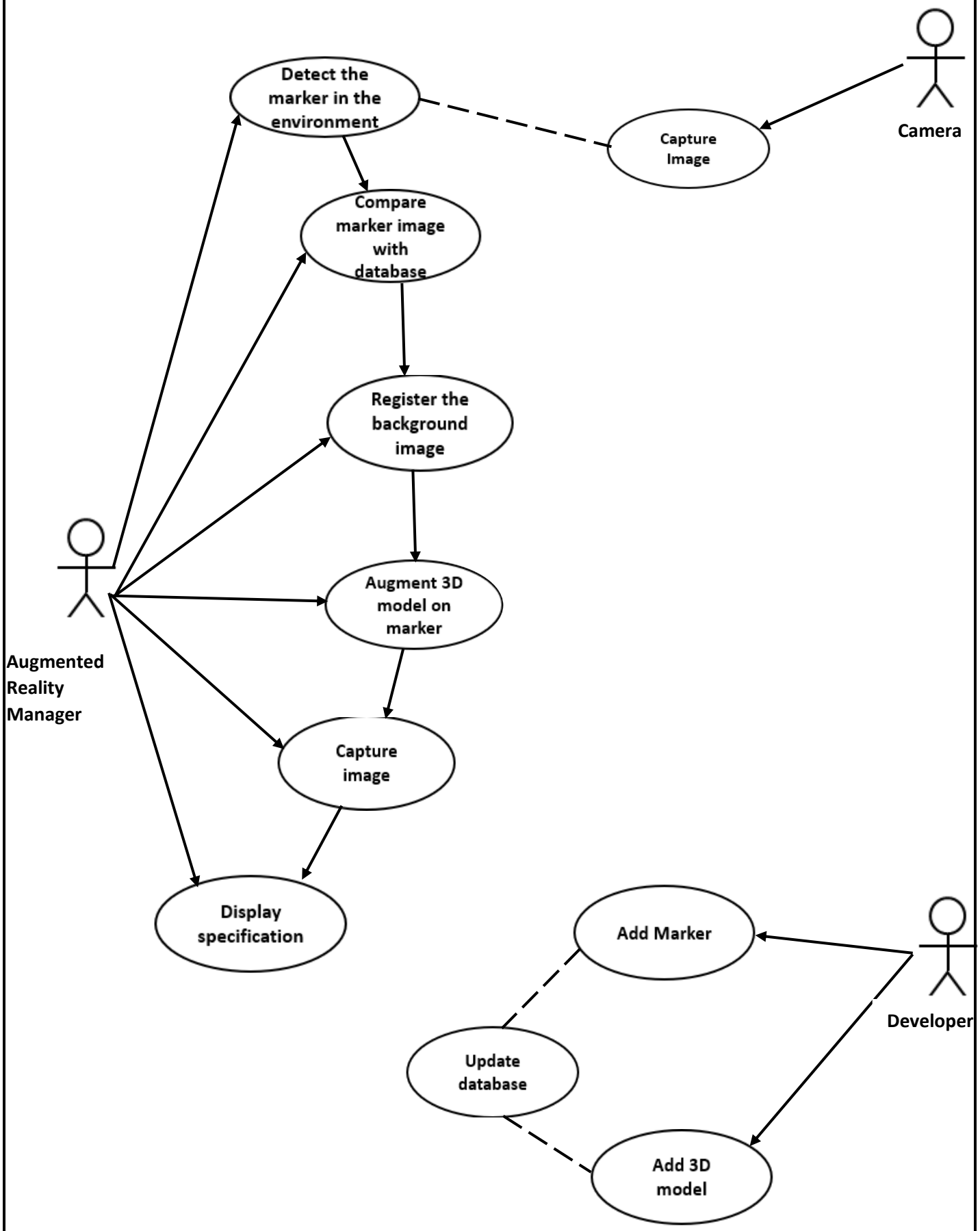


Fig 4.3.1: Use case diagram

- **Activity Diagram:**

Activity diagram is basically a flow chart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. So the control flow is drawn from one operation to another.

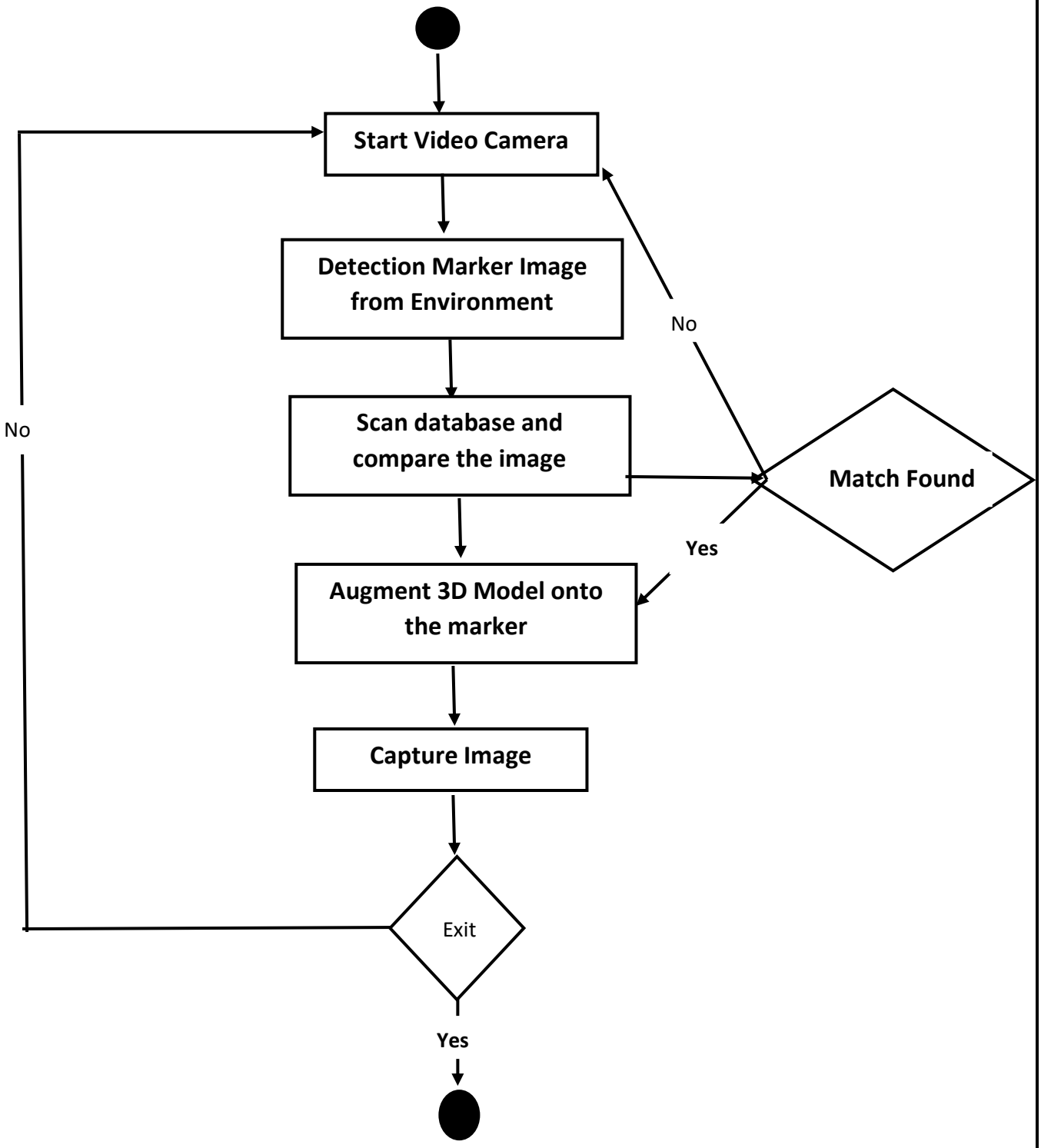


Fig 4.3.2: Activity Diagram

Activity diagram for adding new object:

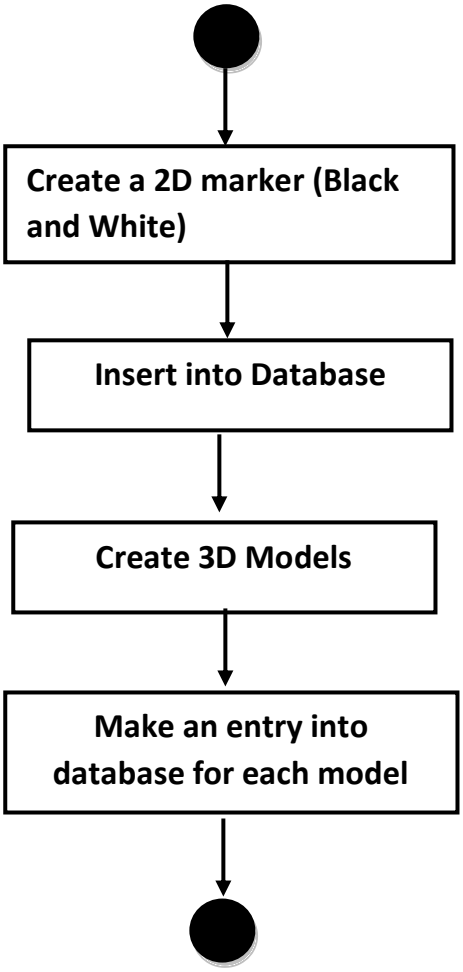


Fig 4.3.3: Activity Diagram for Adding New Object

Explanation:

Step 1: When the customer asks for some furniture model, the Augmented Reality Manager searches for the markers that best fits for the selected model.

Step 2: When the marker is not found, it is required by the developer to create and add 3D model and marker into the data base. It requires following of few necessary steps.

Step-3: We need to create a 2D markers according to the required model demanded by the user. For that we need to create a unique marker, so that the Augmented Reality Manager doesn't get confused.

Step-4: Add it into the database and later add the 3D model into the database.

- **Sequence Diagram:**

A sequence diagram is an interaction diagram that shows how objects operate with one another and in what order. It is a construct of a message sequence chart. A sequence diagram shows object interactions arranged in time sequence.

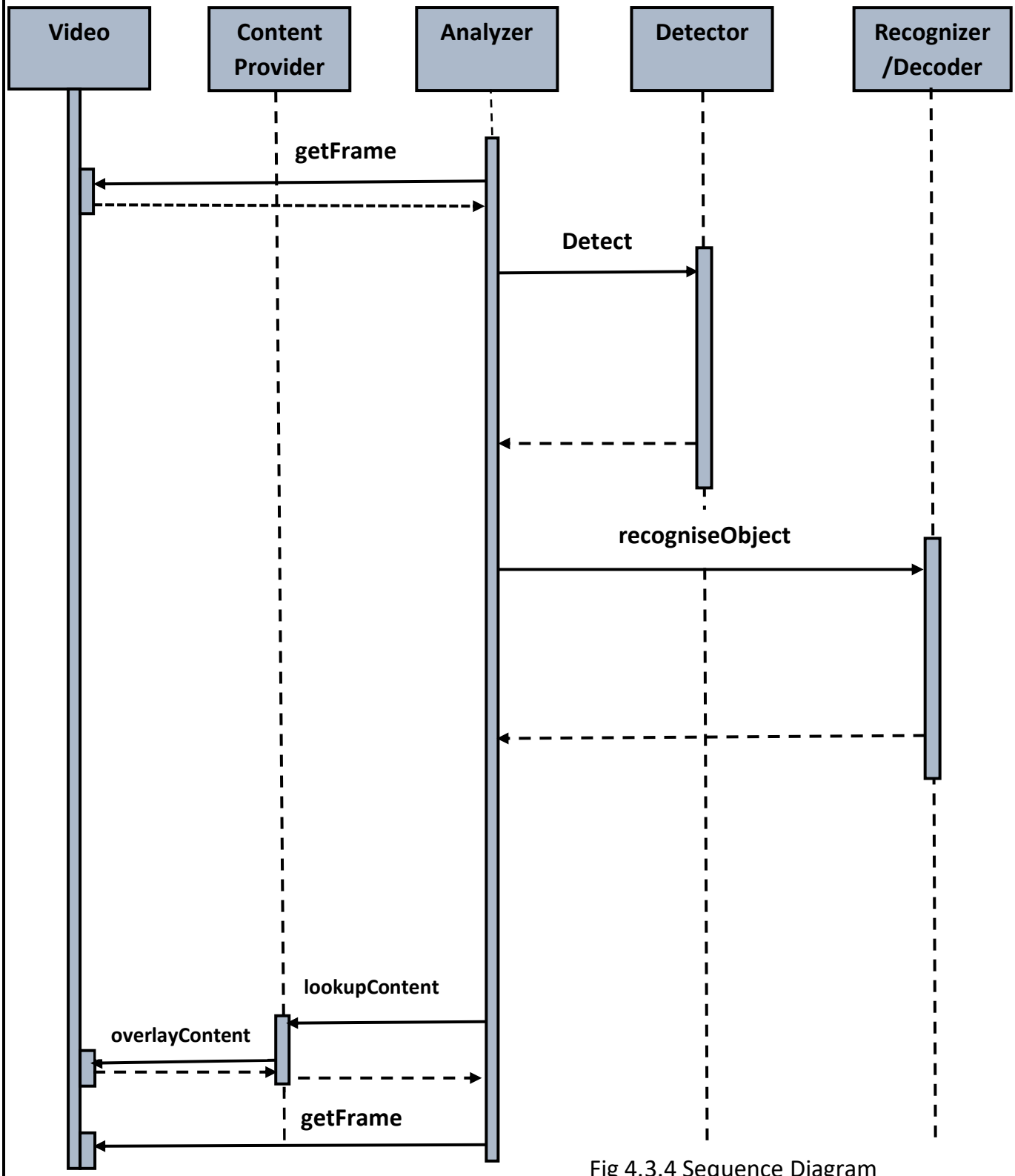


Fig 4.3.4 Sequence Diagram

- **Class Diagram**

A class diagram describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects.

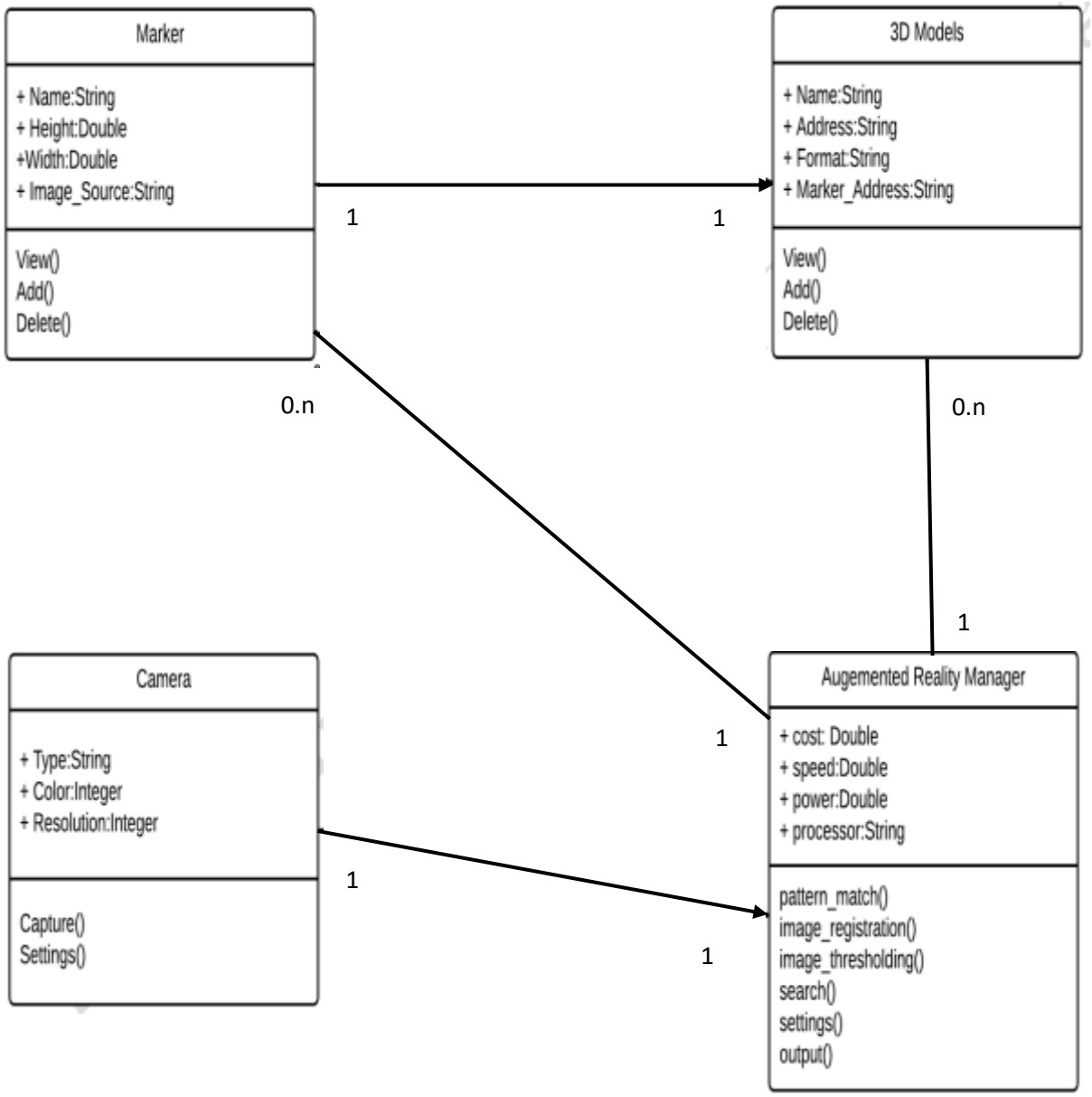


Fig 4.3.5 Class Diagram

- **Flow Chart:**

A graphical representation of the concepts, their principles, elements and components that are part of this flowchart.

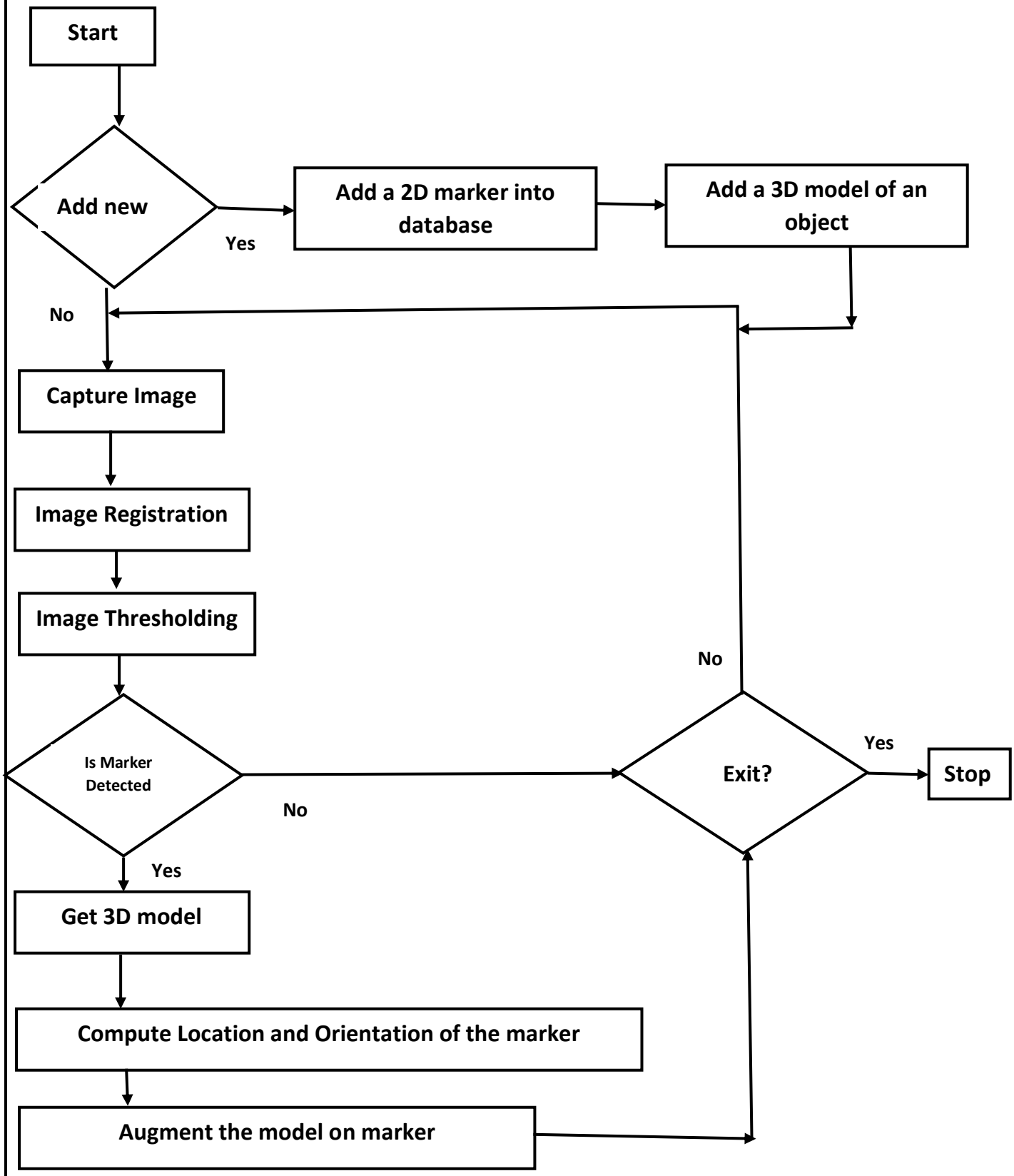


Fig 4.3.6: Flow Chart

5. METHODOLOGY

5.1: Technologies Used

Augmented Reality:

Augmented reality (AR) is an interactive experience of a real-world environment where the objects that reside in the real world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory. AR can be defined as a system that incorporates three basic features: a combination of real and virtual worlds, real-time interaction, and accurate 3D registration of virtual and real objects. The overlaid sensory information can be constructive (i.e. additive to the natural environment), or destructive (i.e. masking of the natural environment). This experience is seamlessly interwoven with the physical world such that it is perceived as an immersive aspect of the real environment. In this way, augmented reality alters one's ongoing perception of a real-world environment, whereas virtual reality completely replaces the user's real-world environment with a simulated one. Augmented reality is related to two largely synonymous terms: mixed reality and computer-mediated reality.

The primary value of augmented reality is the manner in which components of the digital world blend into a person's perception of the real world, not as a simple display of data, but through the integration of immersive sensations, which are perceived as natural parts of an environment. The earliest functional AR systems that provided immersive mixed reality experiences for users were invented in the early 1990s, starting with the Virtual Fixtures system developed at the U.S. Air Force's Armstrong Laboratory in 1992. Commercial augmented reality experiences were first introduced in entertainment and gaming businesses. Subsequently, augmented reality applications have spanned commercial industries such as education, communications, medicine, and entertainment. In education, content may be accessed by scanning or viewing an image with a mobile device or by using marker-less AR techniques.

Augmented reality is used to enhance natural environments or situations and offer perceptually enriched experiences. With the help of advanced AR technologies (e.g. adding computer vision, incorporating AR cameras into smartphone applications and object

recognition) the information about the surrounding real world of the user becomes interactive and digitally manipulated. Information about the environment and its objects is overlaid on the real world. This information can be virtual. Augmented Reality is any experience which is artificial and which adds to the already existing reality. or real, e.g. seeing other real sensed or measured information such as electromagnetic radio waves overlaid in exact alignment with where they actually are in space. Augmented reality also has a lot of potential in the gathering and sharing of tacit knowledge. Augmentation techniques are typically performed in real time and in semantic contexts with environmental elements. Immersive perceptual information is sometimes combined with supplemental information like scores over a live video feed of a sporting event. This combines the benefits of both augmented reality technology and heads up display technology (HUD).

Software and algorithms:

A key measure of AR systems is how realistically they integrate augmentations with the real world. The software must derive real world coordinates, independent of camera, and camera images. That process is called image registration, and uses different methods of computer vision, mostly related to video tracking. Many computer vision methods of augmented reality are inherited from visual odometry. An augogram is a computer-generated image that is used to create AR. Augography is the science and software practice of making augograms for AR.

Usually, those methods consist of two parts. The first stage is to detect interest points, fiducial markers or optical flow in the camera images. This step can use feature detection methods like corner detection, blob detection, edge detection or thresholding, and other image processing methods. The second stage restores a real world coordinate system from the data obtained in the first stage. Some methods assume objects with known geometry (or fiducial markers) are present in the scene. In some of those cases the scene 3D structure should be calculated beforehand. If part of the scene is unknown simultaneous localization and mapping (SLAM) can map relative positions. If no information about scene geometry is available, structure from motion methods like bundle adjustment are used. Mathematical methods used in the second stage include: projective (epipolar) geometry, geometric algebra, rotation representation with exponential map, kalman and particle filters, nonlinear optimization, robust statistics.

In augmented reality, the distinction is made between two distinct modes of tracking, known as marker and marker-less. Markers are visual cues which trigger the display of the virtual information. A piece of paper with some distinct geometries can be used. The camera recognizes the geometries by identifying specific points in the drawing. Marker-less tracking, also called instant tracking, does not use markers. Instead, the user positions the object in the camera view preferably in a horizontal plane. It uses sensors in mobile devices to accurately detect the real-world environment, such as the locations of walls and points of intersection.

Augmented Reality Markup Language (ARML) is a data standard developed within the Open Geospatial Consortium (OGC), which consists of Extensible Markup Language (XML) grammar to describe the location and appearance of virtual objects in the scene, as well as ECMAScript bindings to allow dynamic access to properties of virtual objects.

To enable rapid development of augmented reality applications, some software development applications such as Lens Studio from Snapchat and Spark AR from Facebook were launched including Software Development kits (SDKs) from Apple and Google have emerged.[88][89]

Possible Applications:

Augmented reality has been explored for many applications, from gaming and entertainment to medicine, education and business. Example application areas described below include archaeology, architecture, commerce and education. Some of the earliest cited examples include augmented reality used to support surgery by providing virtual overlays to guide medical practitioners, to AR content for astronomy and welding.

- Archaeology

AR has been used to aid archaeological research. By augmenting archaeological features onto the modern landscape, AR allows archaeologists to formulate possible site configurations from extant structures. Computer generated models of ruins, buildings, landscapes or even ancient people have been recycled into early archaeological AR applications. For example, implementing a system like VITA (Visual Interaction Tool for Archaeology) will allow users to imagine and investigate instant excavation results without leaving their home. Each user can collaborate by mutually "navigating, searching, and viewing data". Hrvoje Benko, a researcher in the computer science department at Columbia University, points out that these particular systems and others

like them can provide "3D panoramic images and 3D models of the site itself at different excavation stages" all the while organizing much of the data in a collaborative way that is easy to use. Collaborative AR systems supply multimodal interactions that combine the real world with virtual images of both environments.

- Architecture**:**

AR can aid in visualizing building projects. Computer-generated images of a structure can be superimposed onto a real-life local view of a property before the physical building is constructed there; this was demonstrated publicly by Trimble Navigation in 2004. AR can also be employed within an architect's workspace, rendering animated 3D visualizations of their 2D drawings. Architecture sight-seeing can be enhanced with AR applications, allowing users viewing a building's exterior to virtually see through its walls, viewing its interior objects and layout.

With continual improvements to GPS accuracy, businesses are able to use augmented reality to visualize georeferenced models of construction sites, underground structures, cables and pipes using mobile devices. Augmented reality is applied to present new projects, to solve on-site construction challenges, and to enhance promotional materials. Examples include the Daqri Smart Helmet, an Android-powered hard hat used to create augmented reality for the industrial worker, including visual instructions, real-time alerts, and 3D mapping.

Following the Christchurch earthquake, the University of Canterbury released CityViewAR, which enabled city planners and engineers to visualize buildings that had been destroyed. This not only provided planners with tools to reference the previous cityscape, but it also served as a reminder of the magnitude of the resulting devastation, as entire buildings had been demolished.

- Urban design and planning**:**

AR systems are being used as collaborative tools for design and planning in the built environment. For example, AR can be used to create augmented reality maps, buildings and data feeds projected onto tabletops for collaborative viewing by built environment professionals. Outdoor AR promises that designs and plans can be superimposed on the real-world, redefining the remit of these professions to bring in-situ design into their process. Design

options can be articulated on site, and appear closer to reality than traditional desktop mechanisms such as 2D maps and 3d models.

Android Studio:

Android Studio is the official integrated development environment (IDE) for Google's Android operating system, built on JetBrains' IntelliJ IDEA software and designed specifically for Android development. It is available for download on Windows, macOS and Linux based operating systems or as a subscription-based service in 2020. It is a replacement for the Eclipse Android Development Tools (E-ADT) as the primary IDE for native Android application development.

Android Studio provides many excellent features that enhance productivity when building Android apps, such as a blended environment where one can develop for all Android devices, apply Changes to push code and resource changes to the running app without restarting the app, a flexible Gradle-based build system, a fast and feature-rich emulator, GitHub and Code template integration to assist you to develop common app features and import sample code, extensive testing tools and frameworks, NDK support, and many more. So we have prepared a complete Android Studio tutorial that will help the Android Developer to get more familiar with Android Studio.

Android Studio was announced on May 16, 2013 at the Google I/O conference. It was in early access preview stage starting from version 0.1 in May 2013, then entered beta stage starting from version 0.8 which was released in June 2014. The first stable build was released in December 2014, starting from version 1.0/. On May 7, 2019, Kotlin replaced Java as Google's preferred language for Android app development. Java is still supported, as is C++. A specific feature of the Android Studio is an absence of the possibility to switch autosave feature off.

The following features are provided in the current stable version:

- Gradle-based build support
- Android-specific refactoring and quick fixes
- Lint tools to catch performance, usability, version compatibility and other problems
- ProGuard integration and app-signing capabilities

- Template-based wizards to create common Android designs and components
- A rich layout editor that allows users to drag-and-drop UI components, option to preview layouts on multiple screen configurations
- Support for building Android Wear apps
- Built-in support for Google Cloud Platform, enabling integration with Firebase Cloud Messaging (Earlier 'Google Cloud Messaging') and Google App Engine
- Android Virtual Device (Emulator) to run and debug apps in the Android studio.

Android Studio supports all the same programming languages of IntelliJ (and CLion) e.g. Java, C++, and more with extensions, such as Go; and Android Studio 3.0 or later supports Kotlin and "all Java 7 language features and a subset of Java 8 language features that vary by platform version." External projects backport some Java 9 features. While IntelliJ states that Android Studio supports all released Java versions, and Java 12, it's not clear to what level Android Studio supports Java versions up to Java 12 (the documentation mentions partial Java 8 support). At least some new language features up to Java 12 are usable in Android.

Once an app has been compiled with Android Studio, it can be published on the Google Play Store. The application has to be in line with the Google Play Store developer content policy.

ARCore:

ARCore is Google’s platform for building augmented reality experiences. Using different APIs, ARCore enables your phone to sense its environment, understand the world and interact with information. Some of the APIs are available across Android and iOS to enable shared AR experiences.

ARCore uses three key capabilities to integrate virtual content with the real world as seen through your phone's camera:

- Motion tracking allows the phone to understand and track its position relative to the world.
- Environmental understanding allows the phone to detect the size and location of all type of

surfaces: horizontal, vertical and angled surfaces like the ground, a coffee table or walls.

- Light estimation allows the phone to estimate the environment's current lighting conditions.

Fundamentally, ARCore is doing two things: tracking the position of the mobile device as it moves, and building its own understanding of the real world.

ARCore's motion tracking technology uses the phone's camera to identify interesting points, called features, and tracks how those points move over time. With a combination of the movement of these points and readings from the phone's inertial sensors, ARCore determines both the position and orientation of the phone as it moves through space.

In addition to identifying key points, ARCore can detect flat surfaces, like a table or the floor, and can also estimate the average lighting in the area around it. These capabilities combine to enable ARCore to build its own understanding of the world around it.

ARCore's understanding of the real world lets you place objects, annotations, or other information in a way that integrates seamlessly with the real world. You can place a napping kitten on the corner of your coffee table, or annotate a painting with biographical information about the artist. Motion tracking means that you can move around and view these objects from any angle, and even if you turn around and leave the room, when you come back, the kitten or annotation will be right where you left it.

ARCore provides SDKs for many of the most popular development environments. These SDKs provide native APIs for all of the essential AR features like motion tracking, environmental understanding, and light estimation. With these capabilities you can build entirely new AR experiences or enhance existing apps with AR features.

Vuforia SDK:

Vuforia is a cross-platform Augmented Reality (AR) and Mixed Reality (MR) application development platform, with robust tracking and performance on a variety of hardware (including mobile devices and mixed reality Head Mounted Displays (HMD) such as the Microsoft HoloLens).

Unity's integration of Vuforia allows you to create vision apps and games for Android and iOS using a drag-and-drop authoring workflow. A Vuforia AR+MR samples package is available on the Unity Asset Store, with several useful examples demonstrating the most important features of the platform.

The Vuforia SDK supports a variety of 2D and 3D target types including 'markerless' Image Targets, 3D Model Target, and a form of addressable Fiducial Marker, known as a VuMark. Additional features of the SDK include 6 degrees of freedom device localization in space, localized Occlusion Detection using 'Virtual Buttons', runtime image target selection, and the ability to create and reconfigure target sets programmatically at runtime.

Vuforia provides Application Programming Interfaces (API) in C++, Java, Objective-C++, and the .NET languages through an extension to the Unity game engine. In this way, the SDK supports both native development for iOS, Android, and UWP while it also enables the development of AR applications in Unity that are easily portable to both platforms.

Vuforia supports many third-party devices (such as AR/MR glasses), VR devices with back-facing cameras (such as the Gear VR). You can use any device with a camera to test AR/MR games and applications built in the Unity with Vuforia.

SQLite:

Sqlite is a database engine written in the C language. It is not a standalone app; rather, it is a library that software developers embed in their apps. As such, it belongs to the family of embedded databases. It is the most widely deployed database engine, as it is used by several of the top web browsers, operating systems, mobile phones, and other embedded systems.

Sqlite has bindings to many programming languages. It generally follows postgresql syntax but does not enforce type checking. This means that one can, for example, insert a string into a column defined as an integer.

Unlike client–server database management systems, the sqlite engine has no standalone processes with which the application program communicates. Instead, the sqlite library is linked in and thus becomes an integral part of the application program. Linking may

be static or dynamic. The application program uses sqlite's functionality through simple function calls, which reduce latency in database access: function calls within a single process are more efficient than inter-process communication.

Sqlite stores the entire database (definitions, tables, indices, and the data itself) as a single cross-platform file on a host machine. It implements this simple design by locking the entire database file during writing. Sqlite read operations can be multitasked, though writes can only be performed sequentially.

Due to the server-less design, sqlite applications require less configuration than client–server databases. Sqlite is called zero-conf because it does not require service management (such as startup scripts) or access control based on GRANT and passwords. Access control is handled by means of file-system permissions given to the database file itself. Databases in client–server systems use file-system permissions that give access to the database files only to the daemon process.

Another implication of the serverless design is that several processes may not be able to write to the database file. In server-based databases, several writers will all connect to the same daemon, which is able to handle its locks internally. Sqlite, on the other hand, has to rely on file-system locks. It has less knowledge of the other processes that are accessing the database at the same time. Therefore, sqlite is not the preferred choice for write-intensive deployments. However, for simple queries with little concurrency, sqlite performance profits from avoiding the overhead of passing its data to another process.

Sqlite uses postgresql as a reference platform. "What would postgresql do" is used to make sense of the SQL standard. One major deviation is that, with the exception of primary keys, sqlite does not enforce type checking; the type of a value is dynamic and not strictly constrained by the schema (although the schema will trigger a conversion when storing, if such a conversion is potentially reversible). Sqlite strives to follow Postel's rule.

Sqlite implements most of the SQL-92 standard for SQL, but lacks some features. For example, it only partially provides triggers and cannot write to views (however, it provides INSTEAD OF triggers that provide this functionality). Its support of ALTER TABLE statements is limited.

Sqlite uses an unusual type system for a SQL-compatible DBMS: instead of assigning a type to a column as in most SQL database systems, types are assigned to individual values; in language

terms it is dynamically typed. Moreover, it is weakly typed in some of the same ways that Perl is: one can insert a string into an integer column (although sqlite will try to convert the string to an integer first, if the column's preferred type is integer). This adds flexibility to columns, especially when bound to a dynamically typed scripting language. However, the technique is not portable to other SQL products. A common criticism is that sqlite's type system lacks the data integrity mechanism provided by statically typed columns in other products. The sqlite web site describes a "strict affinity" mode, but this feature has not yet been added.

Tables normally include a hidden rowid index column, which gives faster access. If a database includes an Integer Primary Key column, sqlite will typically optimize it by treating it as an alias for rowid, causing the contents to be stored as a strictly typed 64-bit signed integer and changing its behavior to be somewhat like an auto-incrementing column. Future Versions of sqlite may include a command to introspect whether a column has behavior like that of rowid to differentiate these columns from weakly typed, non-autoincrementing Integer Primary Keys.

Full support for Unicode case-conversions can be optionally be enabled through an extension.

Several computer processes or threads may access the same database concurrently. Several read accesses can be satisfied in parallel. A write access can only be satisfied if no other accesses are currently being serviced. Otherwise, the write access fails with an error code (or can automatically be retried until a configurable timeout expires). This concurrent access situation would change when dealing with temporary tables. This restriction is relaxed in version 3.7 when write-ahead logging (WAL) is turned on, enabling concurrent reads and writes.

Version 3.6.19 released on October 14, 2009 added support for foreign key constraints.

Sqlite version 3.7.4 first saw the addition of the FTS4 (full-text search) module, which features enhancements over the older FTS3 module. FTS4 allows users to perform full-text searches on documents similar to how search engines search webpages. Version 3.8.2 added support for creating tables without rowid, which may provide space and performance improvements. Common table expressions support was added to sqlite in version 3.8.3. 3.8.11 added a newer search module called FTS5, the more radical (compared to FTS4) changes requiring

a bump in version.

In 2015, with the json1 extension and new subtype interfaces, sqlite version 3.9 introduced JSON content managing.

As of version 3.33.0, the maximum supported database size is 281 TB.

Sqlite's code is hosted with Fossil, a distributed version control system that is itself built upon an sqlite database.

A standalone command-line program is provided in sqlite's distribution. It can be used to create a database, define tables, insert and change rows, run queries and manage an sqlite database file. It also serves as an example for writing applications that use the sqlite library.

Sqlite uses automated regression testing prior to each release. Over 2 million tests are run as part of a release's verification. Starting with the August 10, 2009 release of sqlite 3.6.17, sqlite releases have 100% branch test coverage, one of the components of code coverage. The tests and test harnesses are partially public-domain and partially proprietary.

Java:

Java is a high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible. It is a general-purpose programming language intended to let programmers write once, run anywhere (WORA), meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM) regardless of the underlying computer architecture. The syntax of Java is similar to C and C++, but has fewer low-level facilities than either of them. The Java runtime provides dynamic capabilities (such as reflection and runtime code modification) that are typically not available in traditional compiled languages. As of 2019, Java was one of the most popular programming languages in use according to GitHub, particularly for client–server web applications, with a reported 9 million developers.

Java was originally developed by James Gosling at Sun Microsystems (which has since been acquired by Oracle) and released in 1995 as a core component of Sun Microsystems' Java platform.

The original and reference implementation Java compilers, virtual machines, and class libraries were originally released by Sun under proprietary licenses. As of May 2007, in compliance with the specifications of the Java Community Process, Sun had relicensed most of its Java technologies under the GPL-2.0-only license. Oracle offers its own Hot Spot Java Virtual Machine, however the official reference implementation is the OpenJDK JVM which is free open-source software and used by most developers and is the default JVM for almost all Linux distributions.

As of October 2021, Java 17 is the latest version. Java 8, 11 and 17 are the current long-term support (LTS) versions. Oracle released the last zero-cost public update for the legacy version Java 8 LTS in January 2019 for commercial use, although it will otherwise still support Java 8 with public updates for personal use indefinitely. Other vendors have begun to offer zero-cost builds of OpenJDK 8 and 11 that are still receiving security and other upgrades.

Oracle (and others) highly recommend uninstalling outdated and unsupported versions of Java, because of serious risks due to unresolved security issues. Oracle advises its users to immediately transition to a supported version, such as one of the LTS versions (8, 11, 17).

One design goal of Java is portability, which means that programs written for the Java platform must run similarly on any combination of hardware and operating system with adequate run time support. This is achieved by compiling the Java language code to an intermediate representation called Java bytecode, instead of directly to architecture-specific machine code. Java bytecode instructions are analogous to machine code, but they are intended to be executed by a virtual machine (VM) written specifically for the host hardware. End-users commonly use a Java Runtime Environment (JRE) installed on their device for standalone Java applications or a web browser for Java applets. Standard libraries provide a generic way to access host-specific features such as graphics, threading, and networking. The use of universal bytecode makes porting simple. However, the overhead of interpreting bytecode into machine instructions made interpreted programs almost always run more slowly than native executables. Just-in-time (JIT) compilers that compile byte-codes to machine code during runtime were introduced from an early stage. Java's Hotspot compiler is actually two compilers in one; and with GraalVM (included in e.g. Java 11, but removed as of Java 16) allowing tiered compilation. Java itself is platform-independent and is adapted to the particular platform it is to run on by a Java virtual machine (JVM) for it, which translates the Java bytecode into the platform's machine language.

Programs written in Java have a reputation for being slower and requiring more memory than those written in C++. However, Java programs' execution speed improved significantly with the introduction of just-in-time compilation in 1997/1998 for Java 1.1, the addition of language features supporting better code analysis (such as inner classes, the StringBuilder class, optional assertions, etc.), and optimizations in the Java virtual machine, such as HotSpot becoming Sun's default JVM in 2000. With Java 1.5, the performance was improved with the addition of the java.util.concurrent package, including lock-free implementations of the ConcurrentMaps and other multi-core collections, and it was improved further with Java 1.6.

Some platforms offer direct hardware support for Java; there are micro controllers that can run Java bytecode in hardware instead of a software Java virtual machine, and some ARM-based processors could have hardware support for executing Java bytecode through their Jazelle option, though support has mostly been dropped in current implementations of ARM.

XML:

Extensible Markup Language (XML) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. The design goals of XML focus on simplicity, generality, and usability across the Internet. It is a textual data format with strong support via Unicode for different human languages. Although the design of XML focuses on documents, the language is widely used for the representation of arbitrary data structures such as those used in web services.

The main benefit of xml is that you can use it to take data from a program like Microsoft SQL, convert it into XML then share that XML with other programs and platforms. You can communicate between two platforms which are generally very difficult.

The main thing which makes XML truly powerful is its international acceptance. Many corporation use XML interfaces for databases, programming, office application mobile phones and more. It is due to its platform independent feature.

The design goals of XML emphasize simplicity, generality, and usability across the Internet. It is a textual data format with strong support via Unicode for different human

languages. Although the design of XML focuses on documents, the language is widely used for the representation of arbitrary data structures such as those used in web services.

Several schema systems exist to aid in the definition of XML-based languages, while programmers have developed many application programming interfaces (APIs) to aid the processing of XML data. Many computer systems contain data in incompatible formats. Exchanging data between incompatible systems or upgraded systems is a time-consuming task for web developers. Large amounts of data must be converted, and incompatible data is often lost. XML stores data in plain text format. This provides a software- and hardware-independent way of storing, transporting, and sharing data.

Virtual Emulator:

The Android emulator or Virtual Emulator is an Android Virtual Device (AVD), which represents a specific Android device. We can use the Android emulator as a target device to execute and test our Android application on our PC. The Android emulator provides almost all the functionality of a real device. We can get the incoming phone calls and text messages. It also gives the location of the device and simulates different network speeds. Android emulator simulates rotation and other hardware sensors. It accesses the Google Play store, and much more.

Testing Android applications on emulator are sometimes faster and easier than doing on a real device. For example, we can transfer data faster to the emulator than to a real device connected through USB. The Android emulator comes with predefined configurations for several Android phones, Wear OS, tablet, Android TV devices.

All computers–including mobile devices–work on an ISA, that is, Instruction Set Architecture. This is a set of instructions written in machine-language that your processor understands. Different processor families (think Intel, AMD, ARM, etc.) have their own instruction set architectures, which they implement in their own ways.

The emulator mimics the target/mobile device processor. Then it translates its ISA into the one used by your computer, through a process called binary translation. This binary (ABI–or Application Binary Interface) can be equipped with a compatible operating system and APIs.

The emulator can give you virtual device instances with near-native capabilities and extended controls to adjust the devices' physical sensors, battery state, geolocation, and more.

These near-native capabilities come with a significant performance overhead, mostly due to binary translation.

The best Android Emulators for PC and Mac are:

- BlueStacks
- LDPlayer
- Android Studio
- ARChon
- Bliss OS
- GameLoop
- Genymotion
- MeMU
- MuMu
- Nox
- Phoenix OS
- Prime OS
- Remix OS Player
- Xamarin

5.2 Modules Description:

Module-1: Capture Image Module

Use Case Name	**Captured Image**
Primary Actor	Camera
Goal	To capture live images, this will be used for detection.

Steps	1) Preferably Capture the image in with a plain background like wall, which does not have many colours or texture. This will reduce the time required for marker detection.

Module-2: Add a marker module

Use Case Name	**Add a marker**
Primary Actor	Developer
Goal	In case, if the end user wants new objects to be augmented, then the developer has to create a new marker.
Steps	1) A new 2-D, black and white marker is created and added into the database of existing markers.

Module-3: Add a 3D Model Module

Use Case Name	**Add a marker**
Primary Actor	Developer
Goal	In case, if the end user wants new objects to be augmented, then the developer has to create a new model.
Steps	1) A new 3-D model, similar to the objects to be augmented, then the developer has to create a new model. 2) Add this new model in the database of

	existing model.

Module-4 : Detect a marker module in the environment

User case name	**Detect a marker**
Primary Actor	Augmented Reality Manager
Goal	The system captures a live stream of images in order to compare them with predefined markers.
Steps	1) The first step to detection process is to capture live image using the camera. 2) During this phase, we separate foreground from the background as we know that our marker will be present in the foreground. 3) In order to simplify the above procedure, we make use of black and white markers and thereby first perform image thresholding onto the captured image, to obtain a binary image.

Module-5: Capture marker image with database module

Use case name	**Capture marker image with database module**
Primary Actor	Augmented Reality Manager
Goal	Manager must perform pattern matching to provide a Boolean result i.e. present or not.
Steps	1) The system first extracts the salient features

	of the image like corners, edges, curves etc. which will help in comparison with existing markers. 2) Based on comparison, if the match is above a particular threshold level then true is given as the output else false.

Module-6: Augment 3D model on the marker module

Use case name	**Augment 3D model on the marker**
Primary actor	Augmented Reality Manager
Goal	The model associated with the identified marker is placed on the marker.
Steps	1) Once the marker has been identified, system needs to calculate the location and the orientation of the identified marker. 2) As per the orientation computed, manipulate the models scale and rotation and place the model on computed location.

5.3 Process:

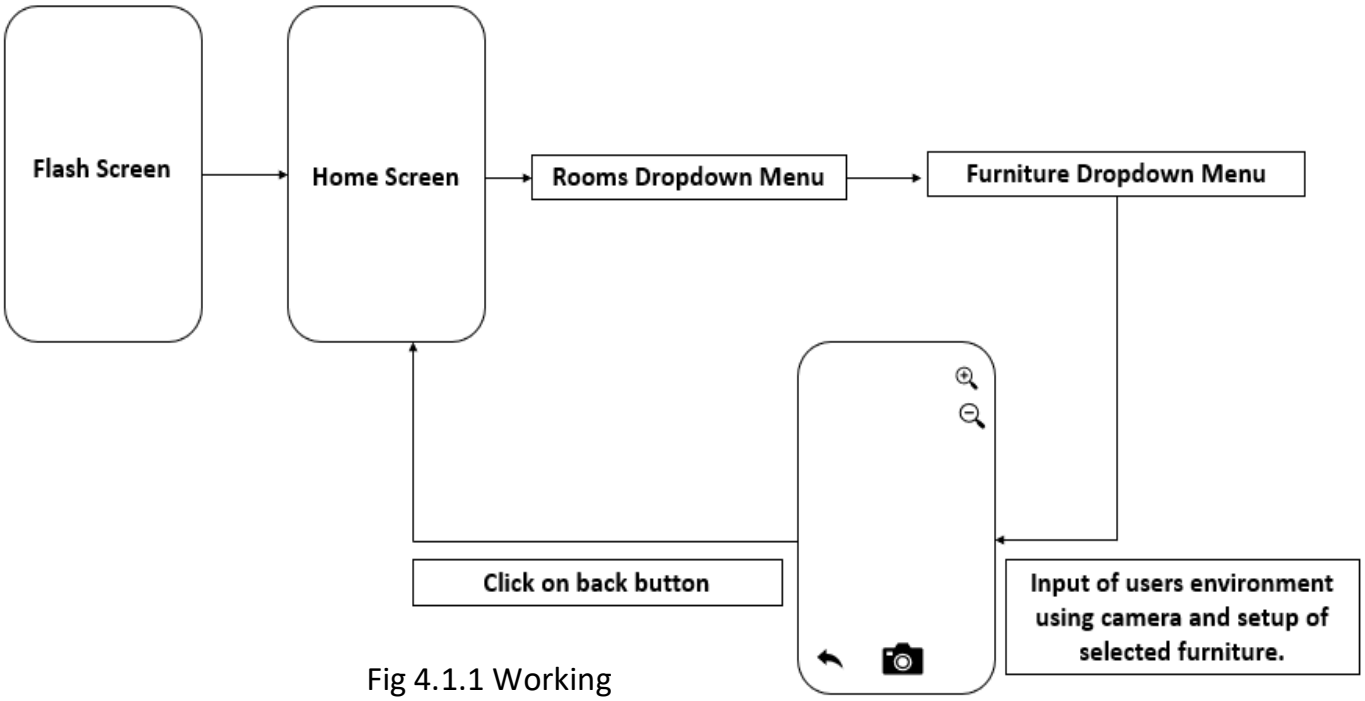


Fig 4.1.1 Working

Step-1: Click on the application which opens the flash screen showing the name of the app for a few fractions of seconds.

Step-2: A login screen appears which shows two options; skip and next. A user can skip and use the app without logging in. But he loses all his data after ending the app.

Step-3: After login, a home screen appears which has a selection menu to select the type of room in the house.

Step-4: After selecting the type of room, there is another selection menu to select the type of furniture to be checked in the selected room.

Step-5: After selecting the furniture type, camera opens. User has to show the environment to the app which marks the points and displays the selected furniture in the space captured.

6. IMPLEMENTATION

6.1: Sample Code

SplashScreen.java

```
package com.furniture.app;

import android.content.Intent;

import android.os.Bundle;

import android.os.Handler;

import androidx.appcompat.app.AppCompatActivity;

import com.furniture.arapp.R;

public class SplashScreen extends AppCompatActivity {

    Handler handler;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.splash_activity);

        handler = new Handler();

        handler.postDelayed(new Runnable() {

            @Override

            public void run() {

                Intent intent = new Intent(SplashScreen.this, LoginActivity.class);

                startActivity(intent);
```

```
            finish();
        }
    }, 3000);
  }
}
```

splash_activity.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<RelativeLayout android:background="@color/purple_700" android:gravity="center"
android:layout_height="match_parent" android:layout_width="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android">

<ImageView android:layout_height="250dp" android:layout_width="250dp"
android:layout_centerInParent="true" android:id="@+id/logo_id"/>

<TextView android:layout_height="wrap_content" android:layout_width="wrap_content"
android:textColor="@color/white" android:textSize="30dp" android:text="Splash Screen"
android:layout_centerHorizontal="true" android:layout_below="@+id/logo_id"/>

</RelativeLayout>
```

RegisterActivity.java:

```
package com.furniture.app;

import android.content.ContentValues;

import android.content.Intent;

import android.database.sqlite.SQLiteDatabase;

import android.os.Bundle;
```

```
import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;

import androidx.annotation.Nullable;

import androidx.appcompat.app.AppCompatActivity;

import com.furniture.app.database.DBHelper;

import com.furniture.arapp.R;

public class RegisterActivity extends AppCompatActivity {

    private EditText ed_email,ed_password,ed_cnPassword;

    private Button btnSignUp;

    private TextView signInTextView;

    @Override

    protected void onCreate(@Nullable Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.register_activity);

        initUI();

        btnSignUp.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View view) {

                if(validate()) {

                    String userName, password;
```

```
            userName = ed_email.getText().toString();

            password = ed_password.getText().toString();

            //cnfPassword = ed.getText().toString();

            // traPassword = transactionPassword.getText().toString();

            DBHelper dbh = new DBHelper(getApplicationContext());

            ContentValues cv = new ContentValues();

            SQLiteDatabase db = dbh.getWritableDatabase();

            cv.put("UserName", userName);

            cv.put("Password", password);

            //cv.put("ConformPassword", cnfPassword);

            //cv.put("TransactionPassword", traPassword);

            db.insert("Details", null, cv);

            Toast.makeText(getApplicationContext(), "success", Toast.LENGTH_LONG).show();

            db.close();

            Intent intent = new Intent(RegisterActivity.this, HomeActivity.class);

            startActivity(intent);

        }

    }

});

signInTextView.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View view) {

        Intent  i = new Intent(RegisterActivity.this,LoginActivity.class);
```

```
            startActivity(i);

        }

    });

  }

  private boolean validate(){

    if(ed_password.getText().length() > 0 && ed_cnPassword.getText().length() > 0){

        Toast.makeText(getApplicationContext(), "Please enter all details",
Toast.LENGTH_LONG).show();

        return  false;

    }else
if(ed_password.getText().toString().trim().equalsIgnoreCase(ed_cnPassword.getText().toString().tri
m())){

        return  true;

    }else{

        Toast.makeText(getApplicationContext(), "Password Not match",
Toast.LENGTH_LONG).show();

        return  false;

    }

  }

  private void initUI() {

    btnSignUp = (Button)  findViewById(R.id.email_sign_Up_button);

    ed_email = (EditText)  findViewById(R.id.email);

    ed_password = (EditText)  findViewById(R.id.password);

    ed_password = (EditText)  findViewById(R.id.cnpassword);

    signInTextView = (TextView)  findViewById(R.id.signINTextView);
```

```
    }

}
```

Register_activity.xml:

```
<?xml version="1.0"?>

<LinearLayout android:focusableInTouchMode="true"
android:descendantFocusability="beforeDescendants" android:background="#ffBE80FF"
android:paddingTop="5dp" android:paddingRight="5dp" android:paddingLeft="5dp"
android:paddingBottom="5dp" android:orientation="vertical"
android:gravity="center_vertical|center_horizontal" android:layout_height="match_parent"
android:layout_width="match_parent" xmlns:tools="http://schemas.android.com/tools"
xmlns:android="http://schemas.android.com/apk/res/android">

<LinearLayout android:gravity="center_vertical|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll1">

<TextView android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/textView2" android:textColor="#ffffffff" android:textSize="30dp"
android:text="Register" android:textAppearance="?android:attr/textAppearanceLarge"/>

</LinearLayout>

<LinearLayout android:gravity="center_vertical|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll2">

<ProgressBar android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/login_progress" android:visibility="gone"
style="?android:attr/progressBarStyleLarge"/>
```

```
<ScrollView android:layout_height="wrap_content" android:layout_width="match_parent"
android:id="@+id/login_form">

<LinearLayout android:orientation="vertical" android:layout_height="wrap_content"
android:layout_width="match_parent">

<LinearLayout android:orientation="vertical" android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email_login_form">

<AutoCompleteTextView android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email" android:textColor="#ffffffff"
android:textColorHint="#ffffffff" android:layout_marginBottom="10dp" android:singleLine="true"
android:maxLines="1" android:inputType="textEmailAddress" android:hint="Enter email"/>

<EditText android:layout_height="wrap_content" android:layout_width="match_parent"
android:id="@+id/password" android:textColor="#ffffffff" android:textColorHint="#ffffffff"
android:layout_marginBottom="10dp" android:singleLine="true" android:maxLines="1"
android:inputType="textPassword" android:hint="Enter password"
android:imeOptions="actionUnspecified"/>

<EditText android:layout_height="wrap_content" android:layout_width="match_parent"
android:id="@+id/cnpassword" android:textColor="#ffffffff" android:textColorHint="#ffffffff"
android:layout_marginBottom="10dp" android:singleLine="true" android:maxLines="1"
android:inputType="textPassword" android:hint="Reenter password"
android:imeOptions="actionUnspecified"/>

<Button android:background="#ff7e51c2" android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email_sign_Up_button"
android:textColor="#ffffffff" android:text="Sign Up" style="?android:textAppearanceSmall"
android:textStyle="bold" android:layout_marginTop="16dp"/>

</LinearLayout>

</LinearLayout>
```

```
</ScrollView>

</LinearLayout>

<LinearLayout android:gravity="bottom|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll3">

<TextView android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/signINTextView" android:textColor="#ffffffff" android:text="Login"
android:autoLink="web"/>

</LinearLayout>

</LinearLayout>
```

LoginActivity.java:

```
package com.furniture.app;

import android.content.Intent;

import android.database.Cursor;

import android.database.sqlite.SQLiteDatabase;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;

import androidx.annotation.Nullable;
```

```
import androidx.appcompat.app.AppCompatActivity;

import com.furniture.app.database.DBHelper;

import com.furniture.arapp.R;

public class LoginActivity extends AppCompatActivity {

    private EditText ed_login,ed_password;

    private Button  btnLogin;

    private TextView signUpTextView;

    @Override

    protected void onCreate(@Nullable Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.login_activty);

        initUI();

        btnLogin.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View view) {

              /* Intent  i = new Intent(LoginActivity.this,MainActivity.class);

               startActivity(i);*/

               doLogin();

            }

        });

        signUpTextView.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View view) {
```

```
            Intent  i = new Intent(LoginActivity.this, RegisterActivity.class);

            startActivity(i);

        }

    });

}

private void initUI() {

    btnLogin = (Button)  findViewById(R.id.email_sign_in_button);

    ed_login = (EditText)  findViewById(R.id.email);

    ed_password = (EditText)  findViewById(R.id.password);

    signUpTextView = (TextView)  findViewById(R.id.signUpTextView);

}

private void doLogin() {

    // TODO Auto-generated method stub

    String Lname,Lpassword;

    Lname = ed_login.getText().toString();

    Lpassword = ed_password.getText().toString();

   /* Intent userDetails = new Intent(MainActivity.this, Home.class);

    startActivity(userDetails);*/

    DBHelper DBCon = new DBHelper(getApplicationContext());

    SQLiteDatabase db = DBCon.getWritableDatabase();

    String sql = "select * from Details where UserName = '"+ Lname+"' and Password =
'"+Lpassword+"'" ;
```

```
 Cursor c = db.rawQuery(sql, null);

      if (c!= null) {

         Intent i = new Intent(LoginActivity.this, HomeActivity.class);

         startActivity(i);

         Toast.makeText(getApplicationContext(), "success",

               Toast.LENGTH_LONG).show();

      } else {

         Toast.makeText(getApplicationContext(), "not success",

               Toast.LENGTH_LONG).show();

      }

   }

}
```

login_activity.xml:

```
<?xml version="1.0"?>

<LinearLayout android:focusableInTouchMode="true"
android:descendantFocusability="beforeDescendants" android:background="#ffBE80FF"
android:paddingTop="5dp" android:paddingRight="5dp" android:paddingLeft="5dp"
android:paddingBottom="5dp" android:orientation="vertical"
android:gravity="center_vertical|center_horizontal" android:layout_height="match_parent"
android:layout_width="match_parent" xmlns:tools="http://schemas.android.com/tools"
xmlns:android="http://schemas.android.com/apk/res/android">

<LinearLayout android:gravity="center_vertical|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll1">
```

```
<TextView android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/textView2" android:textColor="#ffffffff" android:textSize="30dp"
android:text="Login" android:textAppearance="?android:attr/textAppearanceLarge"/>

</LinearLayout>

<LinearLayout android:gravity="center_vertical|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll2">

<ProgressBar android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/login_progress" android:visibility="gone"
style="?android:attr/progressBarStyleLarge"/>

<ScrollView android:layout_height="wrap_content" android:layout_width="match_parent"
android:id="@+id/login_form">

<LinearLayout android:orientation="vertical" android:layout_height="wrap_content"
android:layout_width="match_parent">

<LinearLayout android:orientation="vertical" android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email_login_form">

<AutoCompleteTextView android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email" android:textColor="#ffffffff"
android:textColorHint="#ffffffff" android:layout_marginBottom="10dp" android:singleLine="true"
android:maxLines="1" android:inputType="textEmailAddress" android:hint="Enter Email"/>

<EditText android:layout_height="wrap_content" android:layout_width="match_parent"
android:id="@+id/password" android:textColor="#ffffffff" android:textColorHint="#ffffffff"
android:layout_marginBottom="10dp" android:singleLine="true" android:maxLines="1"
android:inputType="textPassword" android:hint="Enter password"
android:imeOptions="actionUnspecified"/>
```

```
<Button android:background="#ff7e51c2" android:layout_height="wrap_content"
android:layout_width="match_parent" android:id="@+id/email_sign_in_button"
android:textColor="#ffffffff" android:text="signIn" style="?android:textAppearanceSmall"
android:textStyle="bold" android:layout_marginTop="16dp"/>

</LinearLayout>

</LinearLayout>

</ScrollView>

</LinearLayout>

<LinearLayout android:gravity="bottom|center_horizontal"
android:layout_height="match_parent" android:layout_width="match_parent"
android:layout_weight="1" android:id="@+id/ll3">

<TextView android:layout_height="wrap_content" android:layout_width="wrap_content"
android:id="@+id/signUpTextView" android:textColor="#ffffffff" android:text="Sign up"
android:autoLink="web"/>

</LinearLayout>

</LinearLayout>
```

HomeActivity.java:

```
package com.furniture.app;

import android.os.Bundle;

import android.widget.GridView;

import androidx.appcompat.app.AppCompatActivity;

import com.furniture.app.model.Items;
```

```
import com.furniture.arapp.R;

import java.util.ArrayList;

import java.util.List;

public class HomeActivity  extends AppCompatActivity {

   @Override

   protected void onCreate(Bundle savedInstanceState) {

      super.onCreate(savedInstanceState);

      setContentView(R.layout.home_activity);

      List<Items> itemsList = new ArrayList<>();

      itemsList.add(new Items(1,R.drawable.livingroom, "LivingRoom"));

      itemsList.add(new Items(2,R.drawable.kitchen, "Kichen"));

      itemsList.add(new Items(3,R.drawable.bedroom, "BedRoom"));

      itemsList.add(new Items(4,R.drawable.kidsbedroom, "KidsBedRoom"));

      itemsList.add(new Items(5,R.drawable.studyroom, "StudyRoom"));

      GridView gridView = findViewById(R.id.grid_view);

      CustomAdapter customAdapter = new CustomAdapter(this, R.layout.custom_view, itemsList);

      gridView.setAdapter(customAdapter);

   }

}
```

home_activity.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<LinearLayout android:layout_height="match_parent" android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:android="http://schemas.android.com/apk/res/android">

<GridView android:layout_height="match_parent" android:layout_width="match_parent"

app:layout_constraintTop_toTopOf="parent" app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent" app:layout_constraintBottom_toBottomOf="parent"
android:layout_margin="10dp" android:verticalSpacing="10dp" android:numColumns="auto_fit"
android:horizontalSpacing="10dp" android:id="@+id/grid_view"/>

</LinearLayout>
```

CustomAdapter.java

```
package com.furniture.app;

import android.content.Context;

import android.content.Intent;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup;

import android.widget.ArrayAdapter;

import android.widget.ImageView;

import android.widget.LinearLayout;

import android.widget.TextView;

import androidx.annotation.NonNull;

import androidx.annotation.Nullable;
```

```
import com.furniture.app.model.Items;

import com.furniture.arapp.R;

import java.util.ArrayList;

import java.util.List;

public class CustomAdapter extends ArrayAdapter<Items> {

    List<Items> items_list = new ArrayList<>();

    int custom_layout_id;

    private  Context mContext;

public CustomAdapter(@NonNull Context context, int resource, @NonNull List<Items> objects) {

    super(context, resource, objects);

    items_list = objects;

    custom_layout_id = resource;

    mContext= context;

    }

@Override

public int getCount() {

    return items_list.size();

    }

@NonNull

@Override

public View getView(int position, @Nullable View convertView, @NonNull ViewGroup parent) {

    View v = convertView;

    if (v == null) {
```

```
    // getting reference to the main layout and
    // initializing
    LayoutInflater inflater = (LayoutInflater)
getContext().getSystemService(Context.LAYOUT_INFLATER_SERVICE);
    v = inflater.inflate(custom_layout_id, null);
    }
    // initializing the imageview and textview and
    // setting data
    ImageView imageView = v.findViewById(R.id.imageView);
    TextView textView = v.findViewById(R.id.textView);
    LinearLayout itemId = v.findViewById(R.id.itemId);
    // get the item using the  position param
    Items item = items_list.get(position);
    imageView.setImageResource(item.getImage_id());
    textView.setText(item.getText());
    itemId.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent subItem = new Intent(mContext, SubItemsActivity.class);
            subItem.putExtra("ITEMID",item.getItemId());
            mContext.startActivity(subItem);
        }
    });
```

```
        return v;
        }
        }
```

SubItemsActivity.java:

```
package com.furniture.app;
import android.os.Bundle;
import android.widget.GridView;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import com.furniture.app.model.Items;
import com.furniture.arapp.R;
import java.util.List;
public class SubItemsActivity  extends AppCompatActivity {
    private int itemId = 1;
    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.home_activity);
        itemId = getIntent().getIntExtra("ITEMID",1);
        List<Items> itemsList = CommonUtils.getItems(itemId);
        GridView gridView = findViewById(R.id.grid_view);
        SubAdapter customAdapter = new SubAdapter(this, R.layout.custom_view, itemsList);
```

```
        gridView.setAdapter(customAdapter);
    }
}
```

CommonUtils.java

```
package com.furniture.app;
import com.furniture.app.model.Items;
import com.furniture.arapp.R;
import java.util.ArrayList;
import java.util.List;
public class CommonUtils {
    public static List<Items> getItems(int itemId) {
        List<Items> itemsList = new ArrayList<>();
        switch (itemId) {
            case 1:
                itemsList.add(new Items(101, R.drawable.living_room_couch_wo_bg, "couch"));
                itemsList.add(new Items(102, R.drawable.living_room_wardrobe_wo_bg, "wardeobe"));
                break;
            case 2:
                itemsList.add(new Items(201, R.drawable.kitchen_chairs_and_table_wo_bg, "chairs"));
                itemsList.add(new Items(202, R.drawable.kitchen_microwave_wo_bg, "microwave"));
                break;
            case 3:
```

```
            itemsList.add(new Items(301, R.drawable.chair, "chair"));
            itemsList.add(new Items(302, R.drawable.chair, "chair"));
            break;
        case 4:
            itemsList.add(new Items(401, R.drawable.chair, "chair"));
            itemsList.add(new Items(402, R.drawable.chair, "chair"));
            break;
            case 5:
            itemsList.add(new Items(501, R.drawable.chair, "chair"));
            itemsList.add(new Items(402, R.drawable.chair, "chair"));
            itemsList.add(new Items(403, R.drawable.chair, "chair"));
            itemsList.add(new Items(404, R.drawable.chair, "chair"));
            break;
        }
        return itemsList;
    }
}
```

Items.java

```
package com.furniture.app.model;
public class Items {
    private int image_id;
    private int itemId;
```

```
private String text;

public int getImage_id()

{

   return image_id;

}

public void setImage_id(int image_id)

{

   this.image_id = image_id;

}

public String getText()

{

   return text;

}

public int getItemId()

{

   return itemId;

}

public void setText(String text)

{

   this.text = text;

}

public void setItemId(int text)

{
```

```
        this.itemId = itemId;
    }
    public Items(int id,int img, String text)
    {
        image_id = img;
        itemId = id;
        this.text = text;
    }
}
```

DBHandler.java

```
package com.furniture.app.database;
import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
public class DBHelper extends SQLiteOpenHelper{
        public DBHelper(Context context) {
                super(context, "ARTable", null, 3);
                // TODO Auto-generated constructor stub
                }
```

```
        @Override

        public void onCreate(SQLiteDatabase db) {

                // TODO Auto-generated method stub

                db.execSQL(" create table Details( ID INTEGER PRIMARY KEY   AUTOINCREMENT," +

                                "UserName varchar(20),Password varchar(20))");

        }

        @Override

        public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

                // TODO Auto-generated method stub

                //db.execSQL("DROP TABLE IF EXITS History");

                onCreate (db);

        }

}
```

SubAdapter.java

```
package com.furniture.app;

import android.content.Context;

import android.content.Intent;

import android.view.LayoutInflater;

import android.view.View;
```

```
import android.view.ViewGroup;

import android.widget.ArrayAdapter;

import android.widget.ImageView;

import android.widget.LinearLayout;

import android.widget.TextView;

import androidx.annotation.NonNull;

import androidx.annotation.Nullable;

import com.furniture.app.model.Items;

import com.furniture.arapp.R;

import java.util.ArrayList;

import java.util.List;

public class SubAdapter extends ArrayAdapter<Items> {

    List<Items> items_list = new ArrayList<>();

    int custom_layout_id;

    private Context mContext;

    public SubAdapter(@NonNull Context context, int resource, @NonNull List<Items> objects) {

        super(context, resource, objects);

        items_list = objects;

        custom_layout_id = resource;

        mContext= context;

    }

    @Override

    public int getCount() {
```

```
        return items_list.size();

    }

    @NonNull

    @Override

    public View getView(int position, @Nullable View convertView, @NonNull ViewGroup parent) {

        View v = convertView;

        if (v == null) {

            // getting reference to the main layout and

            // initializing

            LayoutInflater inflater = (LayoutInflater)
getContext().getSystemService(Context.LAYOUT_INFLATER_SERVICE);

            v = inflater.inflate(custom_layout_id, null);

        }

        // initializing the imageview and textview and

        // setting data

        ImageView imageView = v.findViewById(R.id.imageView);

        TextView textView = v.findViewById(R.id.textView);

        LinearLayout itemId = v.findViewById(R.id.itemId);

        // get the item using the  position param

        Items item = items_list.get(position);

        imageView.setImageResource(item.getImage_id());

        textView.setText(item.getText());       itemId.setOnClickListener(new View.OnClickListener() {

        itemId.setOnClickListener(new View.OnClickListener() {
```

```
@Override
            public void onClick(View view) {
             Intent subItem = new Intent(mContext, MainActivity.class);
             subItem.putExtra("ITEMID",item.getItemId());
             mContext.startActivity(subItem);
          }
       });
       return v;
   }
}
```

MainActivity.java

```
package com.furniture.app;
import androidx.appcompat.app.AppCompatActivity;
import android.app.Activity;
import android.app.ActivityManager;
import android.app.AlertDialog;
import android.content.Context;
import android.os.Build;
import android.os.Bundle;
import android.view.MotionEvent;
import android.widget.Toast;
import com.google.ar.core.Anchor;
```

```
import com.google.ar.core.HitResult;

import com.google.ar.core.Plane;

import com.google.ar.sceneform.AnchorNode;

import com.google.ar.sceneform.rendering.ModelRenderable;

import com.google.ar.sceneform.rendering.Renderable;

import com.google.ar.sceneform.ux.ArFragment;

import com.google.ar.sceneform.ux.BaseArFragment;

import com.google.ar.sceneform.ux.TransformableNode;

import com.furniture.arapp.R;

import java.lang.ref.WeakReference;

import java.util.Objects;

public class MainActivity extends AppCompatActivity {

    private ArFragment arCam; //object of ArFragment Class

    private   int itemId;

   private int clickNo = 0; //helps to render the 3d models only once when we tap the screen

    public static boolean checkSystemSupport(Activity activity) {

//checking whether the API version of the running Android >= 24 that means Android Nougat 7.0

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.N) {

            String openGlVersion = ((ActivityManager)
Objects.requireNonNull(activity.getSystemService(Context.ACTIVITY_SERVICE))).getDeviceConfigur
ationInfo().getGlEsVersion();

            //checking whether the OpenGL version >= 3.0

            if (Double.parseDouble(openGlVersion) >= 3.0) {

                return true;
```

```
        } else {

            Toast.makeText(activity, "App needs OpenGl Version 3.0 or later",
Toast.LENGTH_SHORT).show();

            activity.finish();

            return false;

        }

    } else {

        Toast.makeText(activity, "App does not support required Build Version",
Toast.LENGTH_SHORT).show();

        activity.finish();

        return false;

    }

  }

  @Override

  protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);

    itemId = getIntent().getExtras().getInt("ITEMID",101);

    if (checkSystemSupport(this)) {

        arCam = (ArFragment)
getSupportFragmentManager().findFragmentById(R.id.arCameraArea);

        //ArFragment is linked up with its respective id used in the activity_main.xml

        arCam.setOnTapArPlaneListener((hitResult, plane, motionEvent) -> {

            clickNo++;
```

```
            //the 3d models comes to the scene only when clickNo is one that means once
            if (clickNo == 1) {
                Anchor anchor = hitResult.createAnchor();
                ModelRenderable.builder()
                        .setSource(this, getItemImage())
                        .setIsFilamentGltf(true)
                        .build()
                        .thenAccept(modelRenderable -> addModel(anchor, modelRenderable))
                        .exceptionally(throwable -> {
                            AlertDialog.Builder builder = new AlertDialog.Builder(this);
                            builder.setMessage("Somthing is not right" + throwable.getMessage()).show();
                            return null;
                        });
                    }
        });
    } else {
        return;
    }
}
private int getItemImage(){
    switch (itemId) {
        case 101:
            return R.raw.livingroomclubchair;
```

```
        case 102:
            return R.raw.livingroomsofa;
        case 201:
            return R.raw.kitchentableandchairs;
        case 202:
            return R.raw.kitchenmicrowave;
        case 301:
            return R.raw.bedroom;
        case 302:
            return R.raw.bedroomsidetable;
        case 401:
            return R.raw.kidsbedroombed;
        case 402:
            return R.raw.kidsbedroomlamp;
        default:
            return R.raw.bedroom;
    }
    }
private void addModel(Anchor anchor, ModelRenderable modelRenderable) {
    AnchorNode anchorNode = new AnchorNode(anchor);
    // Creating a AnchorNode with a specific anchor
    anchorNode.setParent(arCam.getArSceneView().getScene());
    //attaching the anchorNode with the ArFragment
```

```
        TransformableNode model = new TransformableNode(arCam.getTransformationSystem());

        model.setParent(anchorNode);

        //attaching the anchorNode with the TransformableNode

        model.setRenderable(modelRenderable);

        //attaching the 3d models with the TransformableNode that is already attached with the node

        model.select();

    }

}
```

Custom_view.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<LinearLayout android:orientation="vertical" android:background="@drawable/boarder"
android:gravity="center" android:id="@+id/itemId" android:padding="5dp"
android:layout_height="wrap_content" android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools"
xmlns:android="http://schemas.android.com/apk/res/android">

<ImageView android:id="@+id/imageView" android:padding="5dp"
android:layout_height="150dp" android:layout_width="match_parent"
tools:src="@drawable/kitchen"/>

<TextView android:background="@color/teal_700" android:id="@+id/textView"
android:padding="5dp" android:layout_height="wrap_content"
android:layout_width="match_parent" android:textAlignment="center"
android:textColor="@color/white" android:text="TextView"/>

</LinearLayout>
```

ar_layput.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<androidx.constraintlayout.widget.ConstraintLayout tools:context=".MainActivity"
android:layout_height="match_parent" android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:android="http://schemas.android.com/apk/res/android">

<!--This is the fragment that will be used as AR camera-->

<fragment android:layout_height="match_parent" android:layout_width="match_parent"
app:layout_constraintTop_toTopOf="parent" app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent" app:layout_constraintBottom_toBottomOf="parent"
android:name="com.google.ar.sceneform.ux.ArFragment" android:id="@+id/arCameraArea"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Activity_main.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<androidx.constraintlayout.widget.ConstraintLayout tools:context=".MainActivity"
android:layout_height="match_parent" android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:android="http://schemas.android.com/apk/res/android">

<!--This is the fragment that will be used as AR camera-->
```

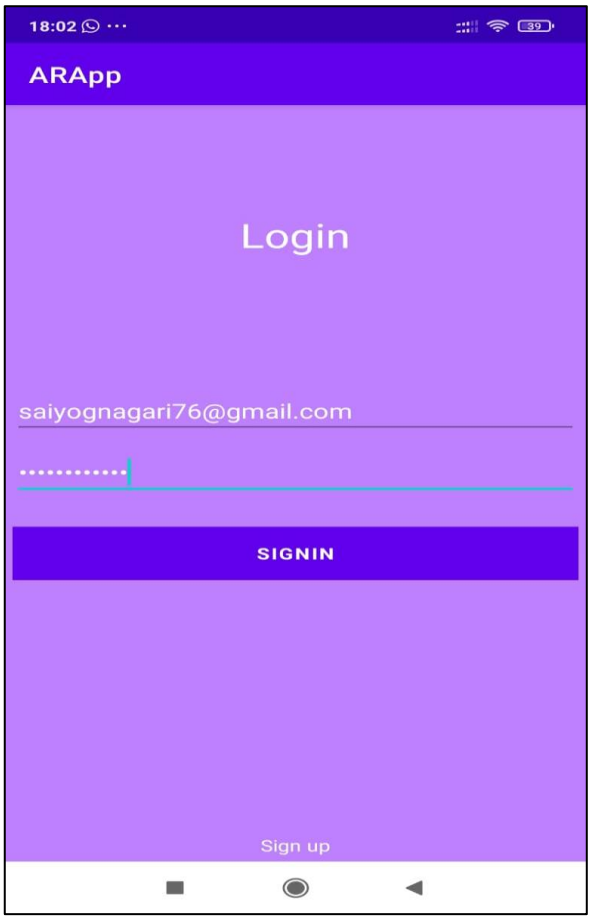
```
<fragment android:layout_height="match_parent" android:layout_width="match_parent"
app:layout_constraintTop_toTopOf="parent" app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent" app:layout_constraintBottom_toBottomOf="parent"
android:name="com.google.ar.sceneform.ux.ArFragment" android:id="@+id/arCameraArea"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

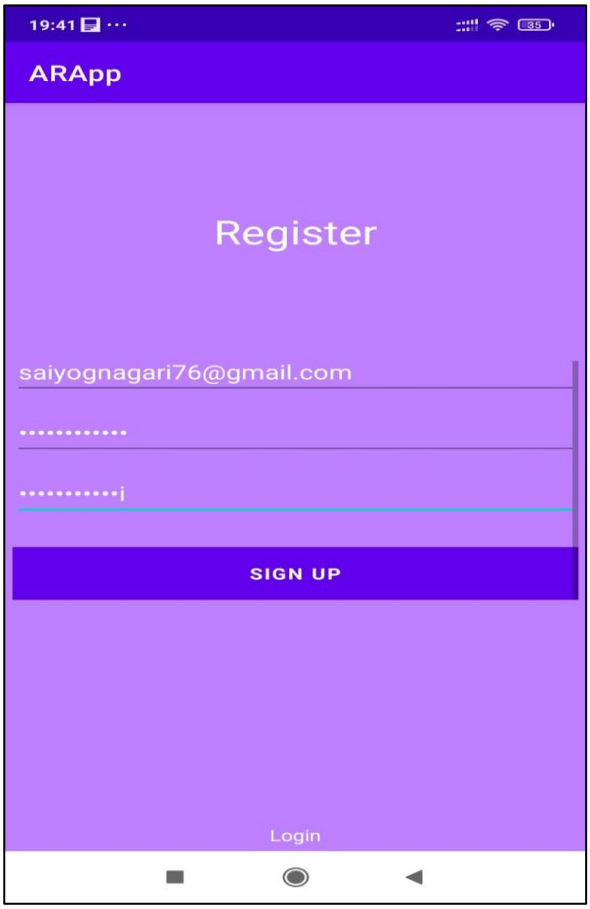
AndroidManifest.xml:

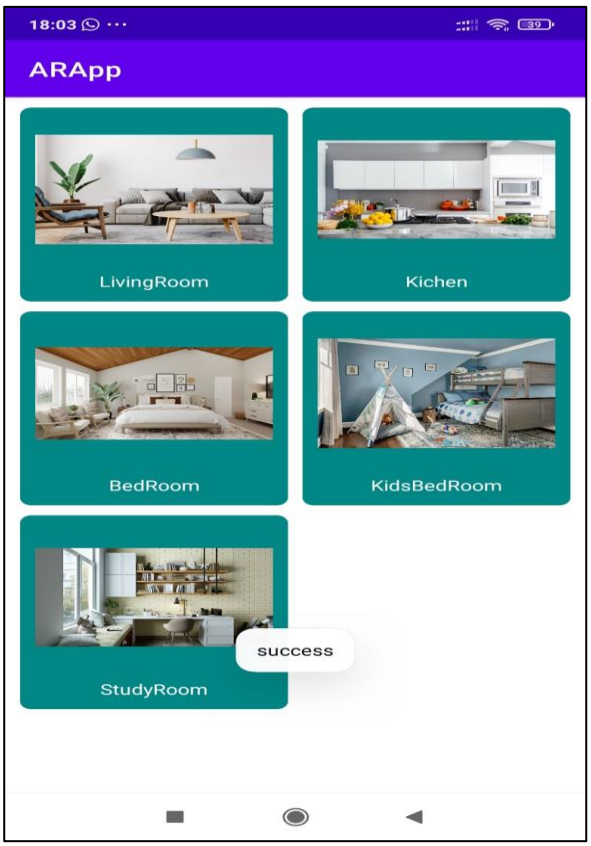
```
<?xml version="1.0" encoding="UTF-8"?>

<manifest package="com.furniture.arapp"
xmlns:android="http://schemas.android.com/apk/res/android">

<uses-permission android:name="android.permission.CAMERA"/>

<!-- This helps to permit the user to access Camera-->

<uses-feature android:required="true" android:glEsVersion="0x00030000"/>

<!-- This helps to check a specific feature in the phone's hardware, here it is OpenGl ES version 3-->

<uses-feature android:name="android.hardware.camera.ar" android:required="true"/>

<!-- Here it is checking for AR feature in phone camera-->

<application android:theme="@style/Theme.ARApp" android:supportsRtl="true"
android:roundIcon="@mipmap/ic_launcher_round" android:label="@string/app_name"
android:icon="@mipmap/ic_launcher" android:allowBackup="true">

<activity android:name="com.furniture.app.SplashScreen"><intent-filter><action
android:name="android.intent.action.MAIN"/>

<category android:name="android.intent.category.LAUNCHER"/></intent-
filter></activity><activity android:name="com.furniture.app.HomeActivity"/>

<activity android:name="com.furniture.app.LoginActivity"/>
```

```
<activity android:name="com.furniture.app.SubItemsActivity"/>

<activity android:name="com.furniture.app.RegisterActivity"/>

<meta-data android:name="com.google.ar.core" android:value="required"/>

<activity android:name="com.furniture.app.MainActivity" android:exported="true">

<intent-filter><action android:name="android.intent.action.MAIN"/>

<category android:name="android.intent.category.DEFAULT"/>

</intent-filter>"></activity></application></manifest>
```

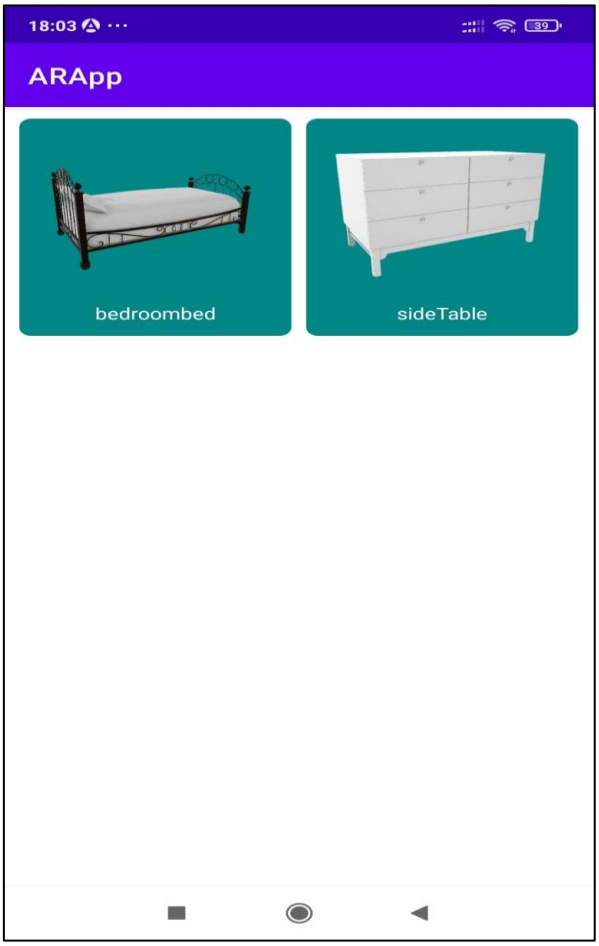
6.2: Output

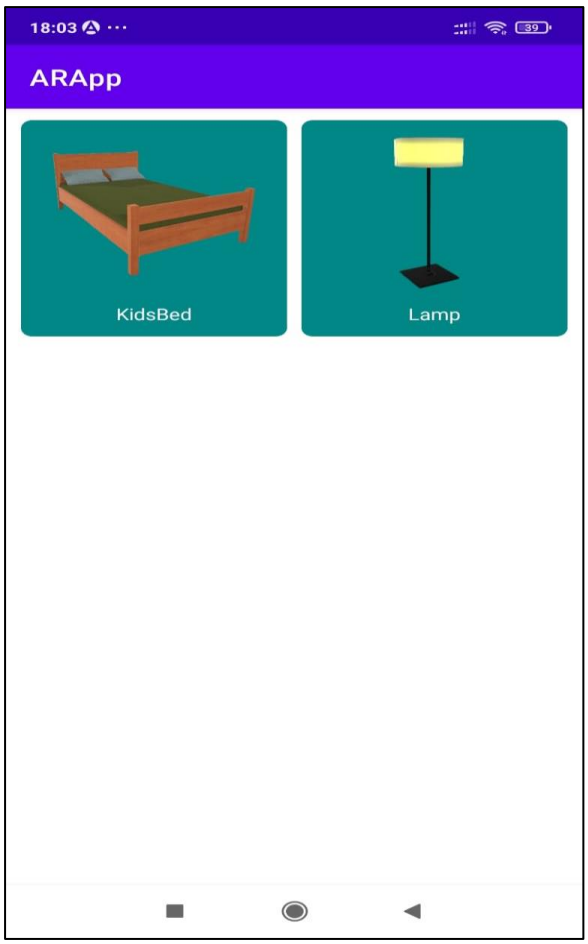

18:02
ARApp
Login
saiyognagari76@gmail.com
SIGNIN
Sign up

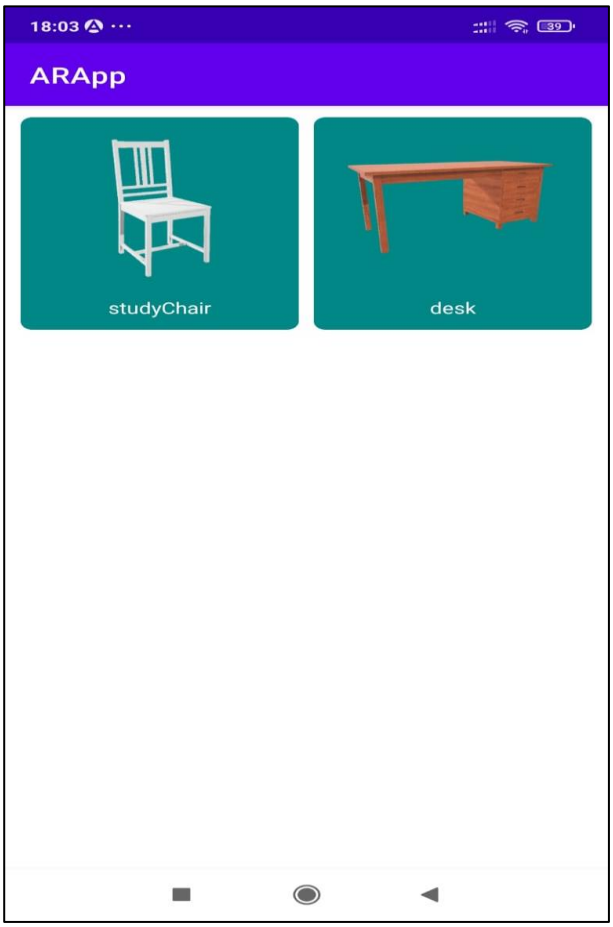

19:41
ARApp
Register
saiyognagari76@gmail.com
SIGN UP
Login


18:03
ARApp
LivingRoom
Kichen
BedRoom
KidsBedRoom
StudyRoom
success

18:03
ARApp
couch
wardeobe


18:03
ARApp
bedroombed
sideTable


18:03
ARApp
KidsBed
Lamp


18:03
ARApp
studyChair
desk

6.3: Test Cases

Login Activity:

Test Case ID	Test Condition	Test Steps	Input Test Data	Expected Result	Pass/Fail
01	Login	Enter phone number and password	Unregistered information	Login unsuccessful	Pass
02	Login	Enter phone number and password	Registered Information	Login successful	Pass

Register Activity:

Test Case ID	Test Condition	Test Steps	Input Test Data	Expected Result	Pass/Fail
01	Register	Enter required fields	Unregistered information	Registration successful	Pass
02	Register	Enter required fields	Registered Information	Registration unsuccessful	Pass

3D Model:

Test Case ID	Test Condition	Test Steps	Input Test Data	Expected Result	Pass/Fail
01	3D object appearance	--	--	3D Object	Pass
02	3D object zoom-in	Zoom-in	--	Large object	Pass
03	3D object zoom-out	Zoom-out	--	Small object	Pass

7. CONCLUSION

7.1 Advantages:

- Application is handy and potable to solve the interior designing task.
- Available to everyone as it is meant for android devices.
- No mandatory chat bot and mandatory signup.
- Can select from wide variety of furniture irrespective of a particular store.
- Time and money friendly.

7.2 Limitations:

- Lesser number of options in furniture.

7.3 Applications:

- Used in e-commerce
- Can be used to extend services of various furniture companies.
- Can be used as individual user to design their homes.

7.4 Future Scope:

- It is possible to integrate the recommendation system so that it becomes easier for the user to purchase the furniture according to desired cost.
- Adding a marker-based AR module for better and easier working of the app.

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